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Towards improving the 2D-MITC4 element for analysis of plane stress and strain problems

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A B S T R A C T

In this paper, we improve the performance of the 2D-MITC4 element for analysis of plane stress and plane strain problems. While the element shows excellent convergence behavior in regular meshes, it exhibits accuracy loss in distorted meshes, as many other elements do. We focus on improving performance in distorted meshes for general use in engineering practice. We simplify the assumed strain field of the original 2D-MITC4 element and introduce the geometry dependent Gauss integration scheme, in which integration points vary according to element geometry distortion. Consequently, a new 2D-MITC4 element is developed. The new 2D-MITC4 element passes the basic tests: patch, zero energy mode, and isotropy tests. Through various numerical examples, improved convergence behaviors of the new element are demonstrated, especially in distorted meshes.

1. Introduction

For the past several decades, the finite element method (FEM) has been widely used for solving various engineering problems [1–3]. FEM can easily handle complex geometries using element meshes. Among various finite elements, low-order finite elements such as 4-node quadrilateral and 3-node triangular elements are often preferred due to simple implementation and computational efficiency compared with higher-order finite elements. However, low-order finite elements show insufficient predictive capability, especially in distorted meshes [4–9].

There have been various approaches used to improve the performance of finite elements, including reduced integration [10–14], assumed strain method, and use of incompatible modes [15,16]. Among them, the method of incompatible modes has been successfully used for low-order finite elements [17–21]; in this method, incompatible modes are adopted to enrich the displacement field of a finite element and corresponding degrees of freedom (DOFs) are added. Regardless of the further computational expense needed to handle the additional DOFs, the incompatible modes element has been widely employed in commercial FE software due to its excellent accuracy improvement in bending problems. However, it reveals spurious instabilities in nonlinear analysis [22–24].

Using the mixed interpolation of tensorial components (MITC) method, the 2D-MITC4 element has been recently developed for two-dimensional analysis of solids [25]. The element does not require additional DOFs, and no numerical instability is observed in nonlinear analysis [26–39]. Through various benchmark problems, the good convergence behavior of the 2D-MITC4 element in regular meshes is shown. However, its performance deteriorates when distorted meshes are used.

The goal of this study is to improve the convergence behavior of the 2D-MITC4 element in distorted meshes. The complicated assumed strain field of the original 2D-MITC4 element is simplified and thus its formulation becomes simpler and more straightforward. We propose a geometry dependent Gauss integration scheme in which integration points are adjusted according to element geometry, leading to an element insensitive to geometry distortion. More specifically, the skewness of the element geometry is measured and integration points are changed according to the degree of skewness. The larger the skewness, the more the integration points move toward the center of the element, by use of an adjusting parameter. We propose an adjusting parameter function to satisfy certain practical requirements. The geometry dependent Gauss integration scheme is very easy to implement and effective in accuracy improvement of the original 2D-MITC4 element. Consequently, we develop the new 2D-MITC4 element for plane stress analysis and the new 2D-MITC4/1 element referring to the new 2D-MITC4 element in which volumetric locking is alleviated for plane stress and plane strain analysis.

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In the following sections, the formulation of the original 2D-MITC4 element is briefly reviewed; then, the formulation of the new 2D-MITC4 element is presented. The performance of the new 2D-MITC4 element is demonstrated through basic numerical tests, and several linear and nonlinear problems considering regular and distorted meshes.

2. Original 2D-MITC4 element

In this section, we briefly review the original formulation of the 2D-MITC4 element including the geometry and displacement interpolations and the assumed strain field [1,25].

2.1. Geometry and displacement interpolations

Considering the natural coordinates r and s, the geometry and displacement of the 4-node quadrilateral 2D element, as shown in Fig. 1, are interpolated by [23]

$$\mathbf{x}(r,s) = \sum_{i=1}^{4} h_i(r,s)\mathbf{x}_i \quad \text{with} \quad \mathbf{x}_i = [\mathrm{x_i}\ \mathrm{y_i}]^T, \tag{1}$$

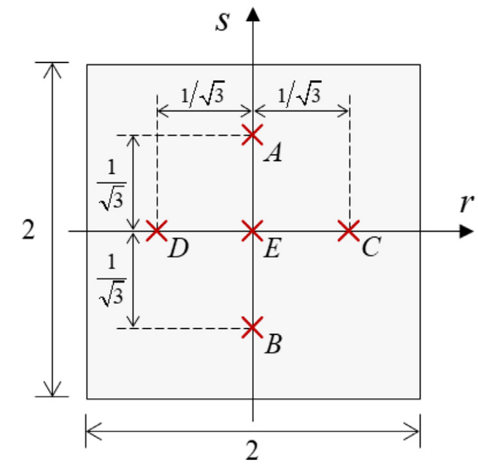

Fig. 3. Five tying points for assumed strain field of the original 2D-MITC4 element.

$$\mathbf{u}(\mathrm{r},\mathrm{s}) = \sum_{i=1}^{4} h_i(r,s)\mathbf{u}_i \quad \text{with} \quad \mathbf{u}_i = [\mathrm{u_i}\ \mathrm{v_i}]^T, \tag{2}$$

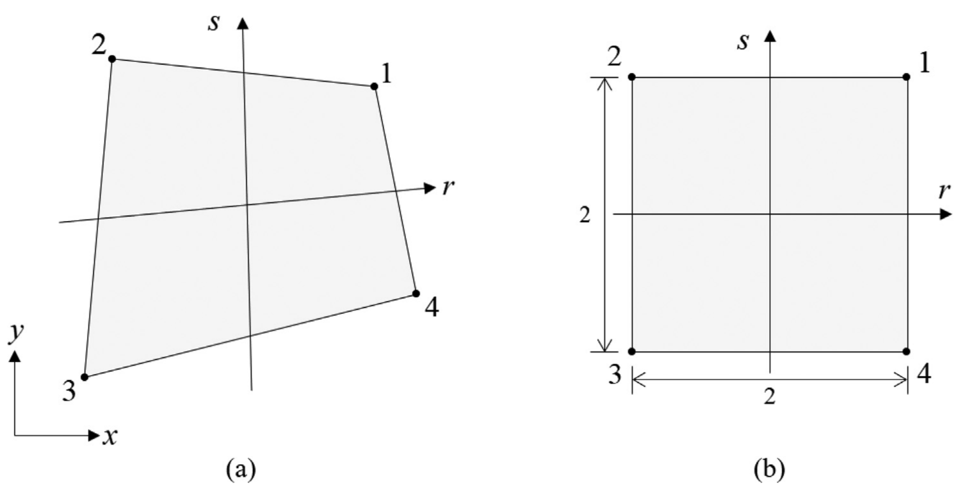

Fig. 1. 4-node quadrilateral element in (a) global Cartesian coordinate system and (b) natural coordinate system.

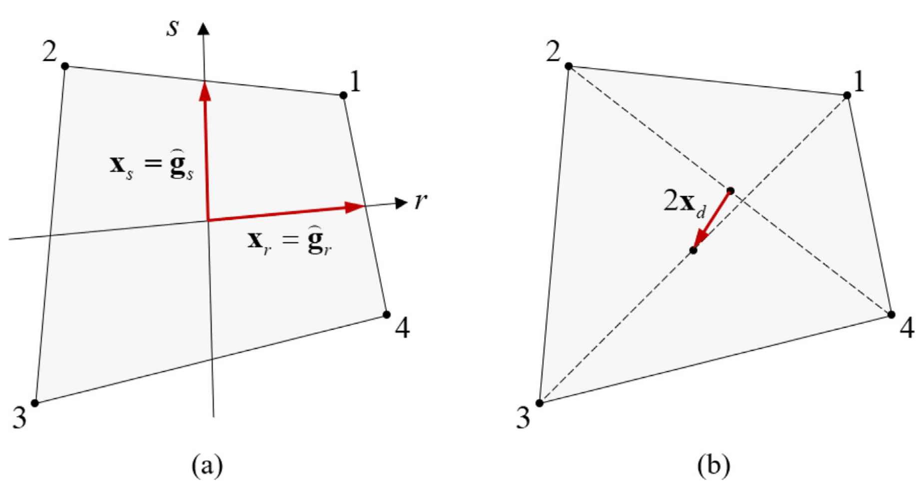

Fig. 2. Characteristic geometry vectors. (a) Vectors $\mathbf{x}_r$ and $\mathbf{x}_s$ correspond to covariant base vectors at element center. (b) Vector $\mathbf{x}_d$ represents in-plane distortion.

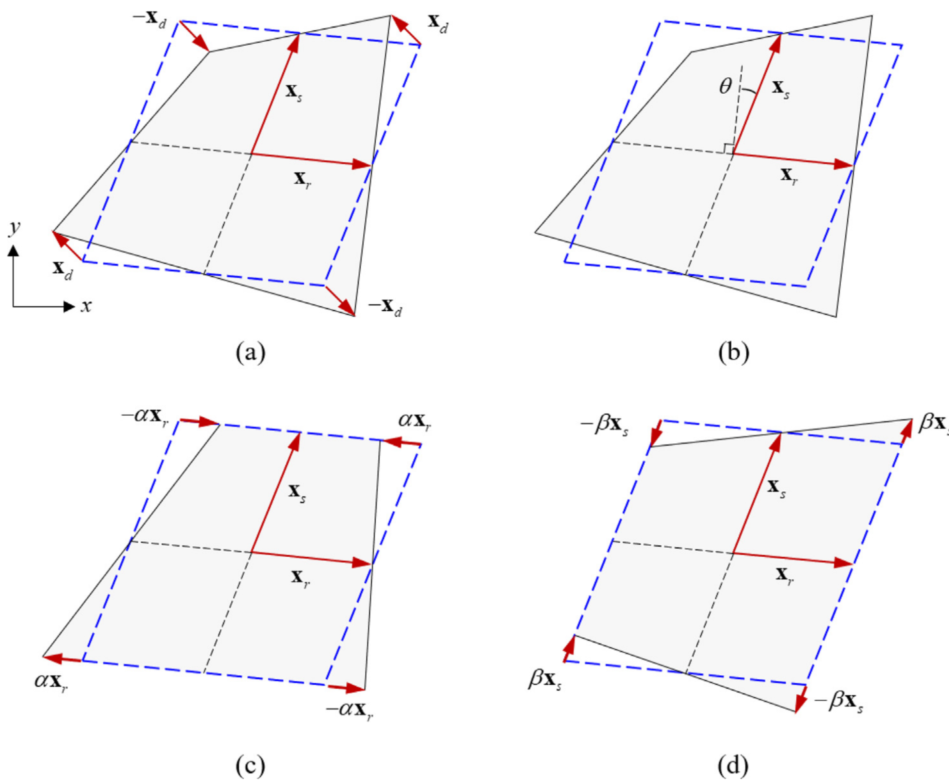

Fig. 4. Classification of distortion for 4-node quadrilateral element. (a) Characteristic vectors, (b) skewness, (c) taper in *r*-direction, and (d) taper in *s*-direction.

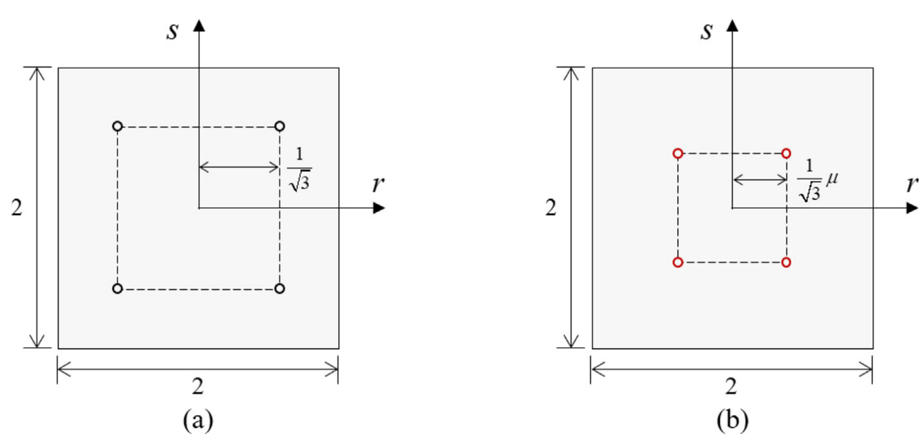

Fig. 5. Integration points for (a) standard Gauss integration and (b) geometry dependent Gauss integration.

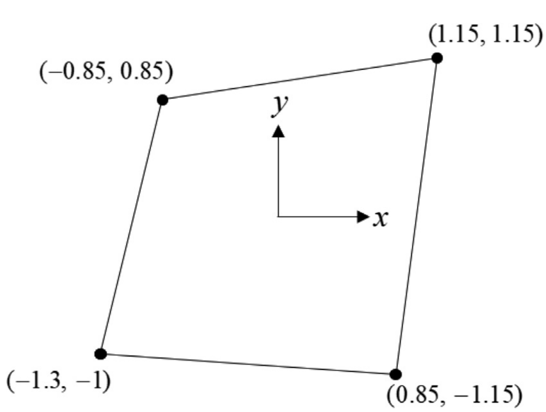

Fig. 6. Distorted element for eigenvalue analysis.

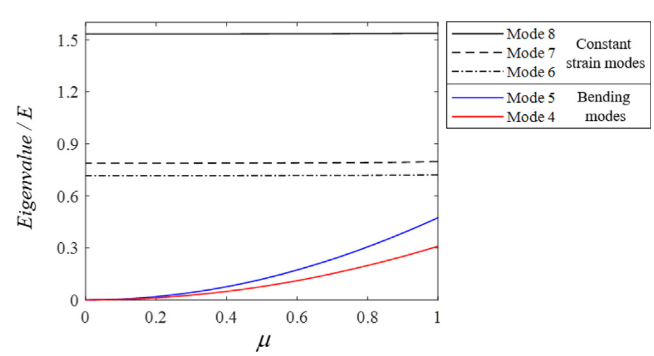

Fig. 7. Change in normalized eigenvalues of each deformation mode according to adjusting parameter μ.

Table 1
Normalized eigenvalues of stiffness matrix of single 2D-MITC4 element according to adjusting parameter μ. The original 2D-MITC4 element corresponds to $\mu = 1$. The eigenvalues are normalized by Young's modulus.

Mode		$\mu = 1$	$\mu = 0.5$	$\mu = 0.1$	$\mu = 0$
1	Rigid body modes	0	0	0	0
2		0	0	0	0
3		0	0	0	0
4	Bending modes	0.3094	0.0773	0.0031	0
5		0.4740	0.1198	0.0048	0
6	Constant strain modes	0.7205	0.7173	0.7169	0.7168
7		0.7976	0.7886	0.7875	0.7874
8		1.5374	1.5339	1.5330	1.5330

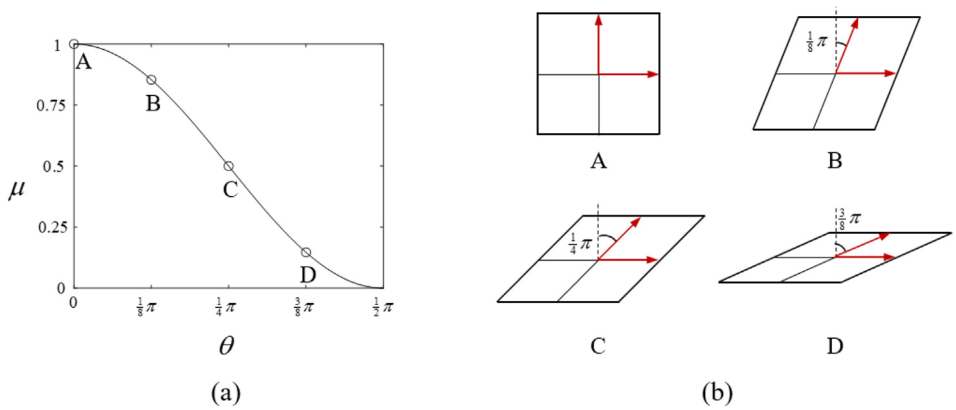

Fig. 8. Adjusting parameter and skew angle. (a) Adjusting parameter according to the skew angle. (b) Element geometries with skew angles $\theta = 0$, $\pi/8$, $\pi/4$, and $3\pi/8$.

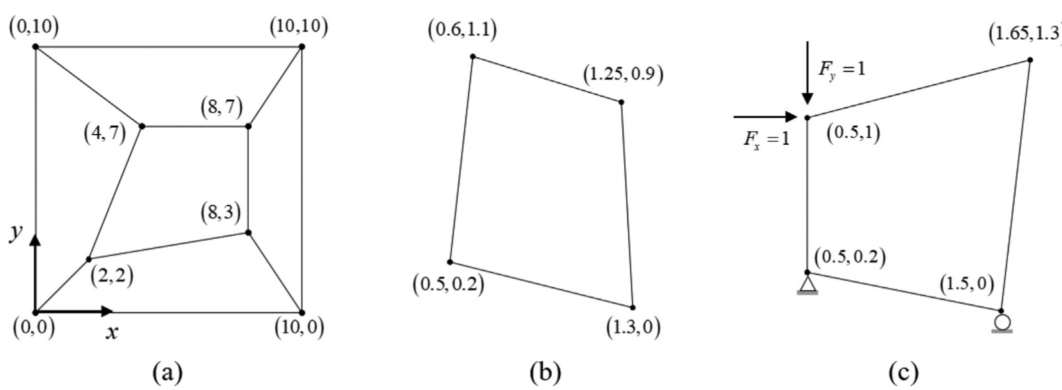

Fig. 9. Geometries for (a) patch, (b) zero energy mode, and (c) isotropy tests.

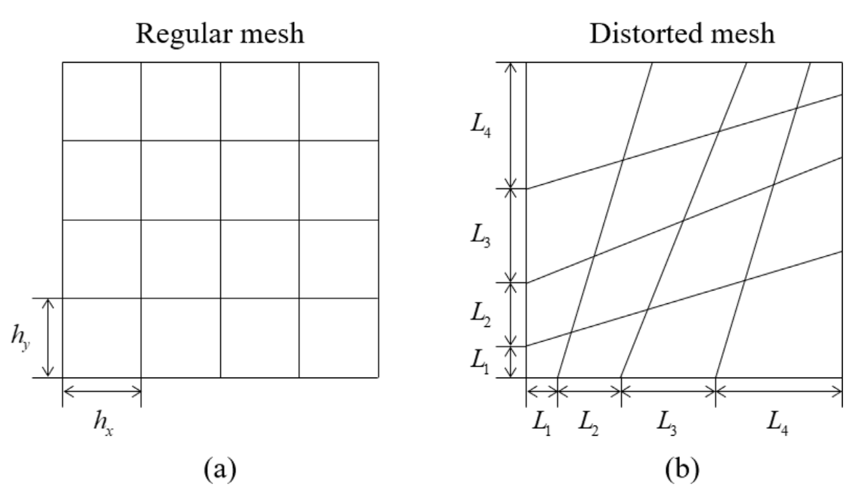

Fig. 10. Mesh patterns when $N = 4$ for (a) regular and (b) distorted meshes.

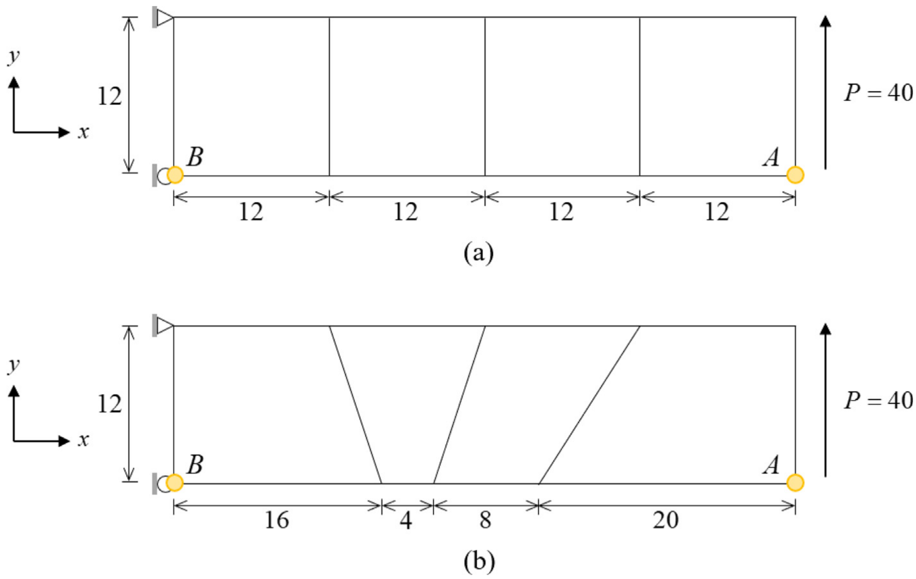

Fig. 11. Cantilever beam ($E = 3.0 \times 10^4$ and $\nu = 0$). (a) Regular mesh. (b) Distorted mesh.

where $\mathbf{x}_i$ is the position vector of node i, $\mathbf{u}_i$ is the displacement vector of node i, and $h_i(r,s)$ is the two-dimensional shape function of the standard isoparametric procedure corresponding to node i,

$$h_i = \frac{1}{4}(1+\xi_i r)(1+\eta_i s) \quad \text{with} \quad i = 1,2,3,4, \tag{3}$$

$$[\,\xi_1 \quad \xi_2 \quad \xi_3 \quad \xi_4\,] = [\,1 \quad -1 \quad -1 \quad 1\,], \tag{4}$$

$$[\,\eta_1 \quad \eta_2 \quad \eta_3 \quad \eta_4\,] = [\,1 \quad 1 \quad -1 \quad -1\,]. \tag{5}$$

The position and displacement vectors can be rewritten as

$$\mathbf{x}(r,s) = \mathbf{x}_a + r\mathbf{x}_r + s\mathbf{x}_s + rs\mathbf{x}_d, \tag{6}$$

$$\mathbf{u}(r,s) = \mathbf{u}_a + r\mathbf{u}_r + s\mathbf{u}_s + rs\mathbf{u}_d, \tag{7}$$

with the following characteristic geometry and displacement vectors [32,46]:

$$\mathbf{x}_a = \frac{1}{4}\sum_{i=1}^{4}\mathbf{x}_i, \quad \mathbf{x}_r = \frac{1}{4}\sum_{i=1}^{4}\xi_i\mathbf{x}_i, \quad \mathbf{x}_s = \frac{1}{4}\sum_{i=1}^{4}\eta_i\mathbf{x}_i, \quad \mathbf{x}_d = \frac{1}{4}\sum_{i=1}^{4}\xi_i\eta_i\mathbf{x}_i, \tag{8}$$

$$\mathbf{u}_a = \frac{1}{4}\sum_{i=1}^{4}\mathbf{u}_i, \quad \mathbf{u}_r = \frac{1}{4}\sum_{i=1}^{4}\xi_i\mathbf{u}_i, \quad \mathbf{u}_s = \frac{1}{4}\sum_{i=1}^{4}\eta_i\mathbf{u}_i, \quad \mathbf{u}_d = \frac{1}{4}\sum_{i=1}^{4}\xi_i\eta_i\mathbf{u}_i, \tag{9}$$

where the vectors in Eq. (8) are determined from the element geometry, shown in Fig. 2.

The covariant base vectors and the derivatives of the displacements are obtained by

$$\mathbf{g}_r = \frac{\partial\mathbf{x}}{\partial r} = \mathbf{x}_r + s\mathbf{x}_d, \quad \mathbf{g}_s = \frac{\partial\mathbf{x}}{\partial s} = \mathbf{x}_s + r\mathbf{x}_d, \tag{10}$$

$$\mathbf{u}_{,r} = \frac{\partial\mathbf{u}}{\partial r} = \mathbf{u}_r + s\mathbf{u}_d, \quad \mathbf{u}_{,s} = \frac{\partial\mathbf{u}}{\partial s} = \mathbf{u}_s + r\mathbf{u}_d. \tag{11}$$

Table 2
Relative errors in tip displacement ($|v_{ref} - v_h|/v_{ref} \times 100$) at point A in cantilever beam.

Element	Regular mesh	Distorted mesh
Q4	32.26 %	41.36 %
ICM-Q4	0 %	1.51 %
2D-MITC4	0 %	13.14 %
New 2D-MITC4	0 %	1.42 %
Reference solution: $v_{ref} = 0.34781$		

Table 3
Relative errors in xx-component of stress ($|\sigma_{ref} - \sigma_h|/\sigma_{ref} \times 100$) at point B in cantilever beam.

Element	Regular mesh	Distorted mesh
Q4	33.33 %	49.80 %
ICM-Q4	0 %	12.07 %
2D-MITC4	0 %	17.03 %
New 2D-MITC4	0 %	12.04 %
Reference solution: $\sigma_{ref} = 70$		

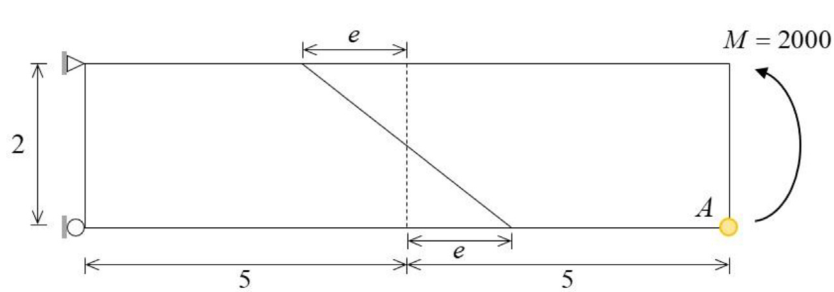

Fig. 12. Cantilever beam ($E = 1500$ and $\nu = 0.25$) modeled using two elements with distortion parameter (e).

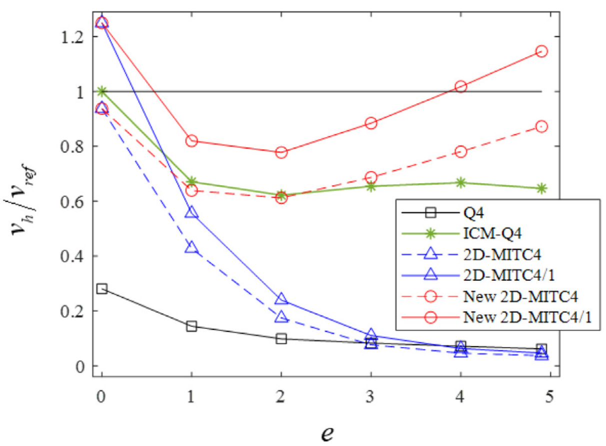

Fig. 13. Normalized vertical displacements (v_h/v_{ref}) at point A according to distortion parameter (e).

Note that the characteristic vectors $\mathbf{x}_r$ and $\mathbf{x}_s$ in Eq. (8) are the covariant base vectors at the element center $(r = s = 0)$,

$$\mathbf{x}_r = \widehat{\mathbf{g}}_r = \mathbf{g}_r(0,0) \quad \text{and} \quad \mathbf{x}_s = \widehat{\mathbf{g}}_s = \mathbf{g}_s(0,0). \tag{12}$$

The covariant strain components are defined by

$$e_{ij} = \frac{1}{2}\left(\mathbf{g}_i \cdot \mathbf{u}_{,j} + \mathbf{g}_j \cdot \mathbf{u}_{,i}\right) \quad \text{with} \quad i,j = 1,2, \tag{13}$$

in which $\mathbf{g}_1 = \mathbf{g}_r$, $\mathbf{g}_2 = \mathbf{g}_s$, $\mathbf{u}_{,1} = \mathbf{u}_{,r}$, and $\mathbf{u}_{,2} = \mathbf{u}_{,s}$. Note that $e_{11} = e_{rr}$, $e_{22} = e_{ss}$ and $e_{12} = e_{rs}$.

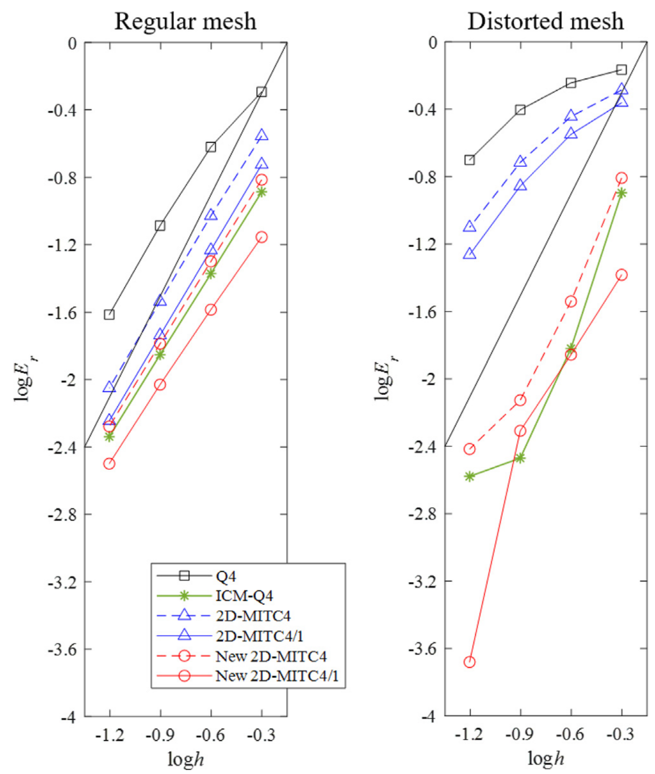

Fig. 15. Convergence curves for Cook's skew beam. The bold line represents the optimal convergence rate.

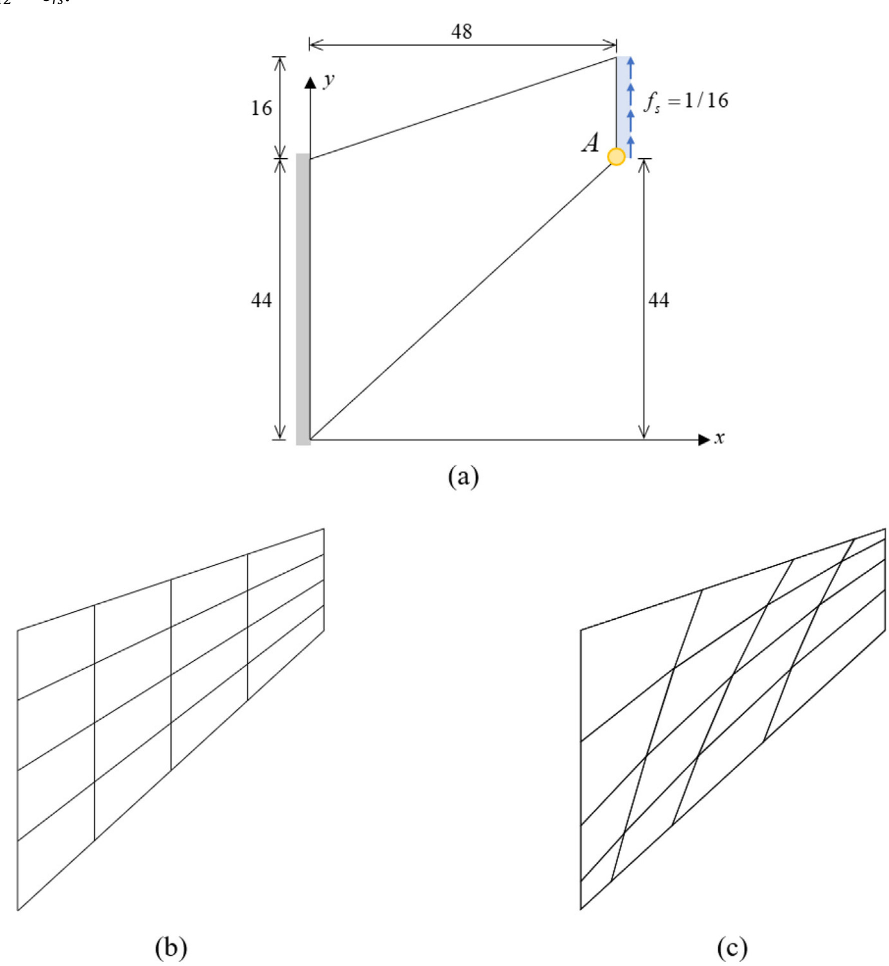

Fig. 14. Problem description of (a) Cook's skew beam ($E = 1.0$ and $\nu = 1/3$). (b) Regular mesh when $N = 4$. (C) Distorted mesh when $N = 4$.

Substituting Eqs. (10) and (11) into Eq. (13), the covariant strain components can be represented using the characteristic vectors in Eqs. (8) and (9), as follows [32]:

$$e_{rr}(r,s) = e_{rr}|_{con} + e_{rr}|_{lin} \cdot s + e_{rs}|_{bil} \cdot s^2, \tag{14}$$

$$e_{ss}(r,s) = e_{ss}|_{con} + e_{ss}|_{lin} \cdot r + e_{rs}|_{bil} \cdot r^2, \tag{15}$$

$$e_{rs}(r,s) = e_{rs}|_{con} + e_{rr}|_{lin} \cdot r + e_{ss}|_{lin} \cdot s + e_{rs}|_{bil} \cdot rs, \tag{16}$$

with the following strain coefficients

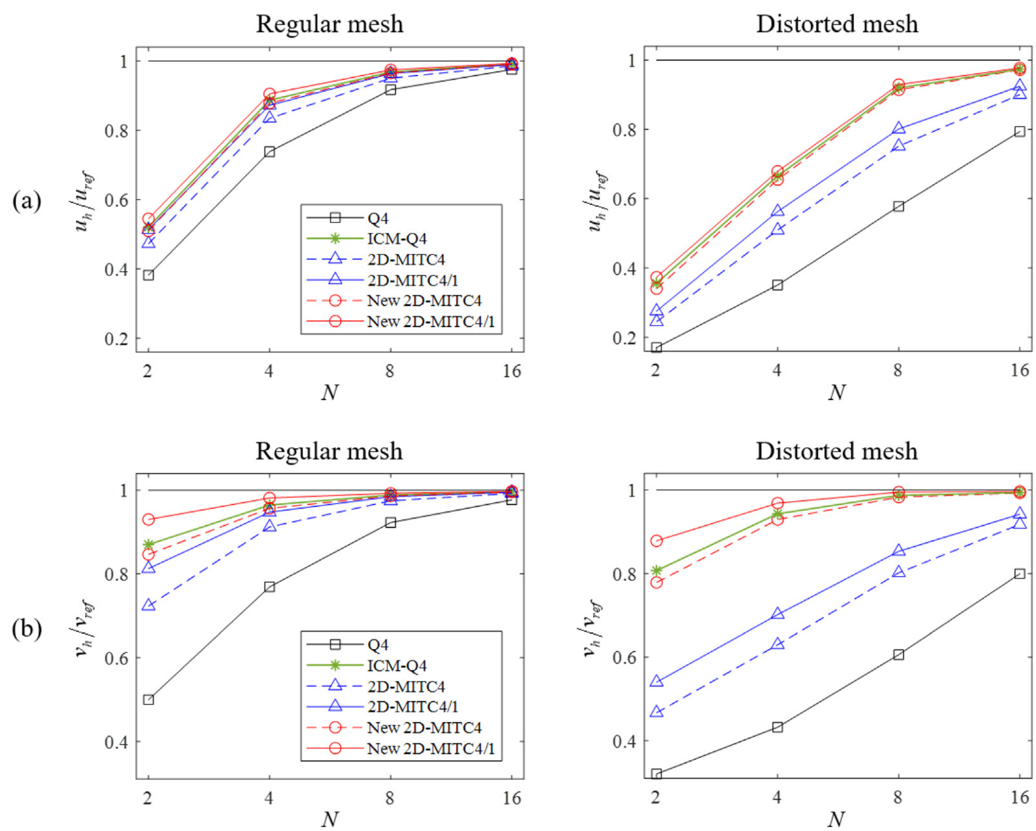

Fig. 16. Normalized (a) horizontal (u_h/u_{ref}) and (b) vertical (v_h/v_{ref}) displacements at point A in Cook's skew beam.

Table 4
Relative errors in horizontal displacement ($|u_{ref} - u_h|/u_{ref} \times 100$) at point A in Cook's skew beam.

	N	Q4	ICM-Q4	2D-MITC4	2D-MITC4/1	New 2D-MITC4	New 2D-MITC4/1
Regular mesh	2	61.79 %	47.98 %	52.72 %	48.70 %	49.04 %	45.54 %
	4	26.18 %	11.31 %	16.55 %	12.78 %	12.17 %	9.49 %
	8	8.28 %	3.17 %	4.95 %	3.63 %	3.47 %	2.56 %
	16	2.45 %	0.90 %	1.44 %	1.02 %	1.00 %	0.71 %
Distorted mesh	2	82.98 %	64.39 %	75.53 %	72.47 %	65.96 %	62.66 %
	4	64.92 %	33.50 %	49.05 %	43.75 %	34.53 %	32.13 %
	8	42.23 %	8.05 %	24.85 %	19.89 %	8.59 %	7.06 %
	16	20.62 %	2.64 %	9.97 %	7.54 %	2.82 %	2.28 %

Reference solution: $u_{ref} = -4.6373$

Table 5
Relative errors in vertical displacement ($|v_{ref} - v_h|/v_{ref} \times 100$) at point A in Cook's skew beam.

	N	Q4	ICM-Q4	2D-MITC4	2D-MITC4/1	New 2D-MITC4	New 2D-MITC4/1
Regular mesh	2	50.06 %	13.00 %	27.69 %	18.74 %	15.35 %	6.99 %
	4	23.11 %	3.55 %	8.79 %	5.25 %	4.33 %	1.87 %
	8	7.75 %	1.17 %	2.55 %	1.55 %	1.38 %	0.74 %
	16	2.31 %	0.38 %	0.77 %	0.49 %	0.45 %	0.26 %
Distorted mesh	2	67.99 %	19.29 %	53.31 %	45.99 %	22.14 %	12.17 %
	4	56.77 %	5.69 %	37.04 %	29.83 %	7.06 %	3.13 %
	8	39.41 %	1.28 %	19.83 %	14.68 %	1.70 %	0.48 %
	16	20.05 %	0.58 %	8.22 %	5.78 %	0.70 %	0.34 %

Reference solution: $v_{ref} = 23.2022$

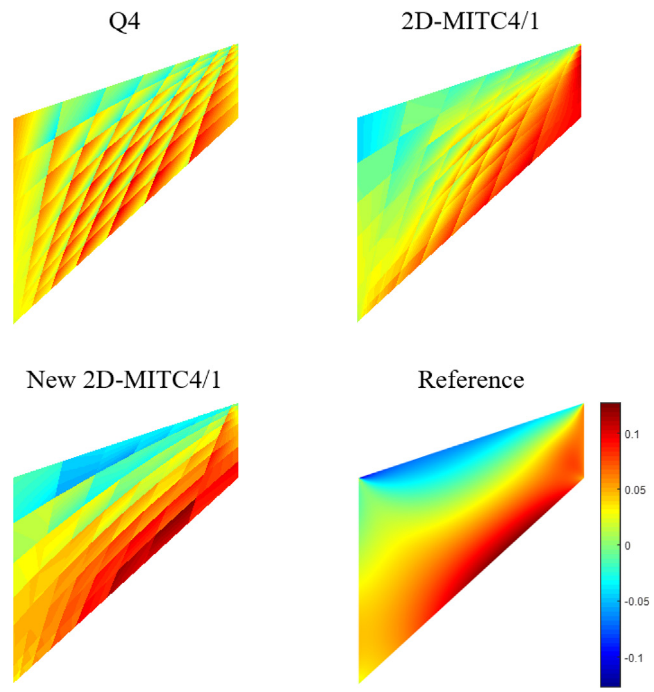

Fig. 17. Shear stress (σ_{xy}) distributions calculated in Cook's skew beam using distorted mesh with $N = 8$. The reference stress distribution is obtained using a 64×64 mesh of 9-node quadrilateral elements.

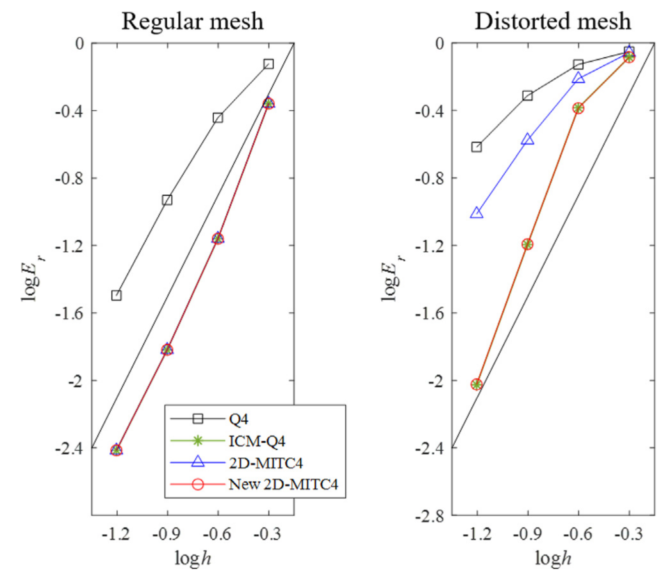

Fig. 19. Convergence curves for curved beam. The bold line denotes the optimal convergence rate.

$$e_{rr}|_{con} = \mathbf{x}_r \cdot \mathbf{u}_r, \; e_{rr}|_{lin} = \mathbf{x}_r \cdot \mathbf{u}_d + \mathbf{x}_d \cdot \mathbf{u}_r, \tag{17}$$

$$e_{ss}|_{con} = \mathbf{x}_s \cdot \mathbf{u}_s, \; e_{ss}|_{lin} = \mathbf{x}_s \cdot \mathbf{u}_d + \mathbf{x}_d \cdot \mathbf{u}_s, \tag{18}$$

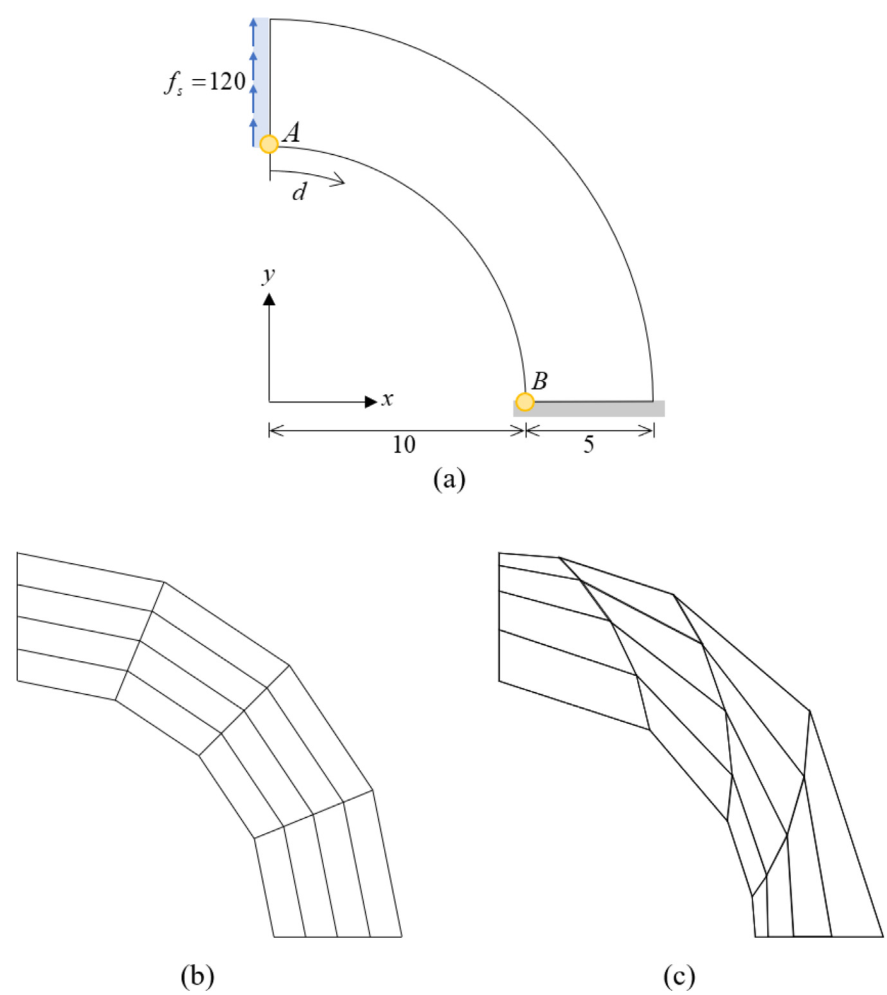

Fig. 18. Problem description of (a) curved beam ($E = 1.0 \times 10^3$ and $\nu = 0$). (b) Regular mesh when $N = 4$. (c) Distorted mesh when $N = 4$.

$$e_{rs}|_{con} = \frac{1}{2}(\mathbf{x}_r \cdot \mathbf{u}_s + \mathbf{x}_s \cdot \mathbf{u}_r), \; e_{rs}|_{bil} = \mathbf{x}_d \cdot \mathbf{u}_d, \tag{19}$$

in which the subscripts '*con*', '*lin*', and '*bil*' denote constant, linear, and bilinear terms of the strain components, respectively.

2.2. Assumed strain field

In the original 2D-MITC4 element, the assumed strain field is constructed for the following strain components

$$\hat{e}_{ij} = e_{kl} g_i^k g_j^l, , , \text{with} \quad \mathbf{g}^i \cdot \hat{\mathbf{g}}_j = g_j^i, \tag{20}$$

where $\mathbf{g}^i$ is the contravariant base vector and $\hat{\mathbf{g}}_i$ denotes the covariant base vectors defined in Eq. (12).

Employing five tying points (*A*)-(*E*) in Fig. 3, the assumed strain field is given as [25]:

$$\hat{e}_{rr}^{AS}(r,s) = \hat{e}_{rr}^{(E)} + \frac{\sqrt{3}}{2}\lambda(r,s)\left(\hat{e}_{rr}^{(A)} - \hat{e}_{rr}^{(B)}\right)s, \tag{21}$$

$$\hat{e}_{ss}^{AS}(r,s) = \hat{e}_{ss}^{(E)} + \frac{\sqrt{3}}{2}\lambda(r,s)\left(\hat{e}_{ss}^{(C)} - \hat{e}_{ss}^{(D)}\right)r, \tag{22}$$

$$\hat{e}_{rs}^{AS}(r,s) = \hat{e}_{rs}^{(E)}, \tag{23}$$

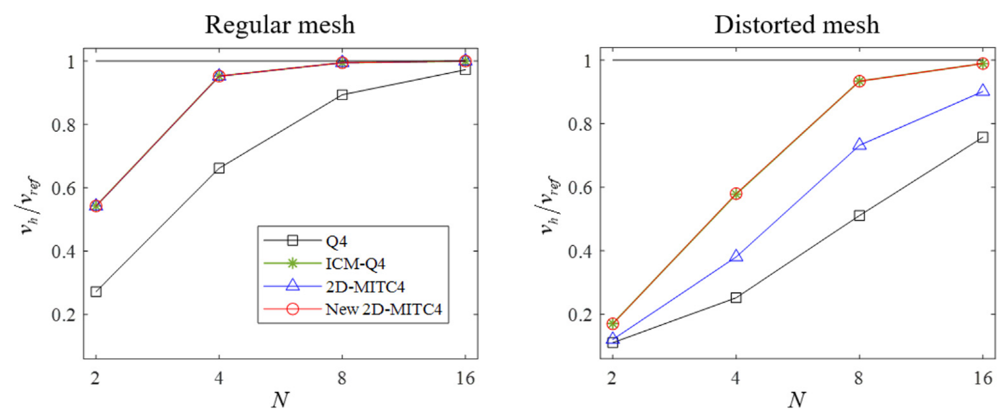

Fig. 20. Normalized vertical displacements (v_h/v_{ref}) at point *A* in curved beam.

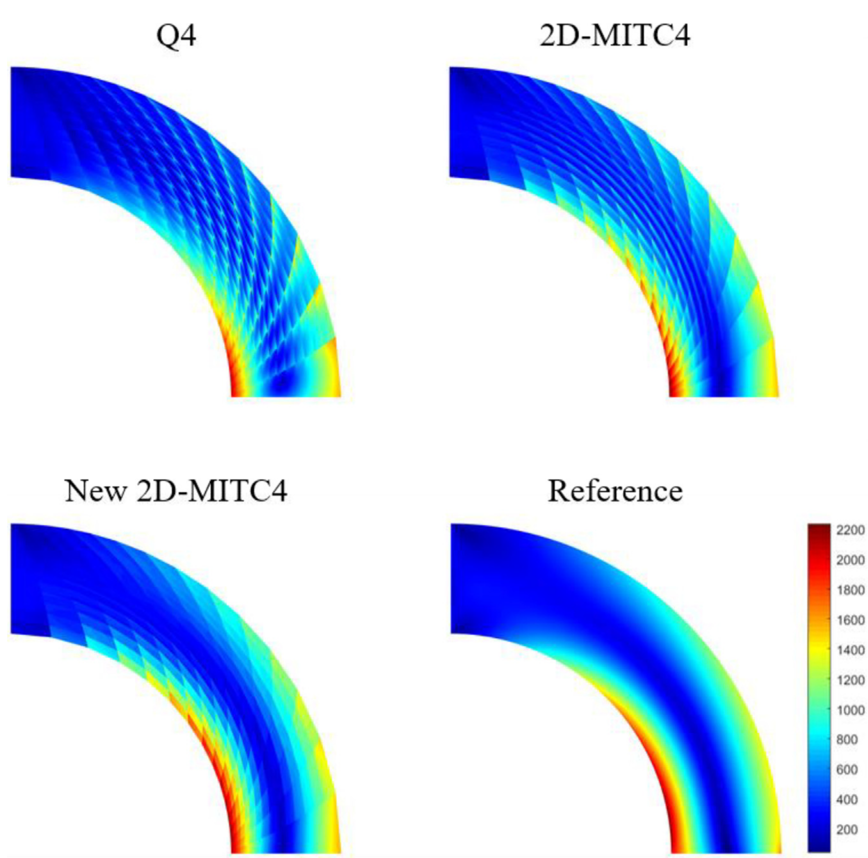

Fig. 21. Von mises stress (τ_{vM}) distributions of curved beam obtained using 16 × 16 distorted mesh. The reference stress distribution is obtained using a 64 × 64 mesh of 9-node quadrilateral elements.

with

$$\lambda(r,s) = \frac{\det(\boldsymbol{J}(0,0))}{\det(\boldsymbol{J}(r,s))}, \tag{24}$$

where $\boldsymbol{J}$ denotes the Jacobian matrix, λ is the ratio of the determinants of the Jacobian matrices, and the superscript '*AS*' denotes the assumed strain component.

To effectively calculate the strain field in Eqs. (21)-(23) without performing the strain transformation in Eq. (20), the assumed strain field of the original 2D-MITC4 element is represented by employing the strain coefficients in Eqs. (17)-(19)

$$\overset{\frown}{e}{}_{rr}^{AS}(r,s) = e_{rr}|_{con} + \lambda(r,s)\hat{e}_{rr}^{lin}s, \tag{25}$$

$$\overset{\frown}{e}{}_{ss}^{AS}(r,s) = e_{ss}|_{con} + \lambda(r,s)\hat{e}_{ss}^{lin}r, \tag{26}$$

$$\overset{\frown}{e}{}_{rs}^{AS}(r,s) = e_{rs}|_{con}, \tag{27}$$

with

$$\hat{e}_{rr}^{lin} = \frac{n_1}{\sqrt{3}}e_{rs}|_{bil} + \sqrt{3}n_1 e_{rr}|_{con} + \sqrt{3}n_2 e_{ss}|_{con} + n_3 e_{rr}|_{lin} + n_4 e_{ss}|_{lin} + 2\sqrt{3}n_5 e_{rs}|_{con}, \tag{28}$$

$$\hat{e}_{ss}^{lin} = \frac{m_1}{\sqrt{3}}e_{rs}|_{bil} + \sqrt{3}m_1 e_{ss}|_{con} + \sqrt{3}m_2 e_{rr}|_{con} + m_3 e_{ss}|_{lin} + m_4 e_{rr}|_{lin} + 2\sqrt{3}m_5 e_{rs}|_{con}, \tag{29}$$

$$n_1 = \frac{1}{2}\left[\left(g_r^r|_{(A)}\right)^2 - \left(g_r^r|_{(B)}\right)^2\right],$$
$$n_2 = \frac{1}{2}\left[\left(g_r^s|_{(A)}\right)^2 - \left(g_r^s|_{(B)}\right)^2\right], n_3 = \frac{1}{2}\left[\left(g_r^r|_{(A)}\right)^2 + \left(g_r^r|_{(B)}\right)^2\right], \tag{30}$$

$$n_4 = \frac{1}{2}\left[g_r^r|_{(A)} \cdot g_r^s|_{(A)} + g_r^r|_{(B)} \cdot g_r^s|_{(B)}\right],$$
$$n_5 = \frac{1}{2}\left[g_r^r|_{(A)} \cdot g_r^s|_{(A)} - g_r^r|_{(B)} \cdot g_r^s|_{(B)}\right], \tag{31}$$

$$m_1 = \frac{1}{2}\left[\left(g_s^s|_{(C)}\right)^2 - \left(g_s^s|_{(D)}\right)^2\right],$$
$$m_2 = \frac{1}{2}\left[\left(g_s^r|_{(C)}\right)^2 - \left(g_s^r|_{(D)}\right)^2\right],$$
$$m_3 = \frac{1}{2}\left[\left(g_s^s|_{(C)}\right)^2 + \left(g_s^s|_{(D)}\right)^2\right], \tag{32}$$

$$m_4 = \frac{1}{2}\left[g_s^r|_{(C)} \cdot g_s^s|_{(C)} + g_s^r|_{(D)} \cdot g_s^s|_{(D)}\right],$$
$$m_5 = \frac{1}{2}\left[g_s^r|_{(C)} \cdot g_s^s|_{(C)} - g_s^r|_{(D)} \cdot g_s^s|_{(D)}\right], \tag{33}$$

in which $g_i^j\big|_{(\cdot)}$ is g_i^j evaluated at a tying point $(\cdot)$.

2.3. Simplification of assumed strain field

The strain components at each tying point are represented using the characteristic geometry and displacement vectors in Eqs. (8)-(9), and the resulting equations are substituted into Eqs. (21)-(23); the assumed strain field can be rewritten as

$$\overset{\frown}{e}{}_{rr}^{AS}(r,s) = \mathbf{x}_r \cdot \mathbf{u}_r + \frac{3\lambda(r,s)}{3-\alpha^2}(-\alpha\mathbf{x}_r \cdot \mathbf{u}_r - \beta\mathbf{x}_r \cdot \mathbf{u}_s + \mathbf{x}_r \cdot \mathbf{u}_d)s, \tag{34}$$

$$\overset{\frown}{e}{}_{ss}^{AS}(r,s) = \mathbf{x}_s \cdot \mathbf{u}_s + \frac{3\lambda(r,s)}{3-\beta^2}(-\beta\mathbf{x}_s \cdot \mathbf{u}_s - \alpha\mathbf{x}_s \cdot \mathbf{u}_r + \mathbf{x}_s \cdot \mathbf{u}_d)r, \tag{35}$$

$$\overset{\frown}{e}{}_{rs}^{AS}(r,s) = \frac{1}{2}(\mathbf{x}_r \cdot \mathbf{u}_s + \mathbf{x}_s \cdot \mathbf{u}_r), \tag{36}$$

with

$$\alpha = \mathbf{x}_d \cdot \overset{\frown}{\mathbf{g}}{}^r \text{ and } \beta = \mathbf{x}_d \cdot \overset{\frown}{\mathbf{g}}{}^s, \tag{37}$$

where the vectors $\overset{\frown}{\mathbf{g}}{}^r$ and $\overset{\frown}{\mathbf{g}}{}^s$ are the contravariant base vectors evaluated at the element center with $\overset{\frown}{\mathbf{g}}{}^r = \mathbf{g}^r(0,0)$ and $\overset{\frown}{\mathbf{g}}{}^s = \mathbf{g}^s(0,0)$ as in Eq. (12).

The simplified assumed strain field in Eqs. (34)-(37) directly consists of the characteristic vectors. This assumed strain field has a much simpler form than that of the original assumed strain field, which consists of the strain coefficients in Eqs. (25)-(33).

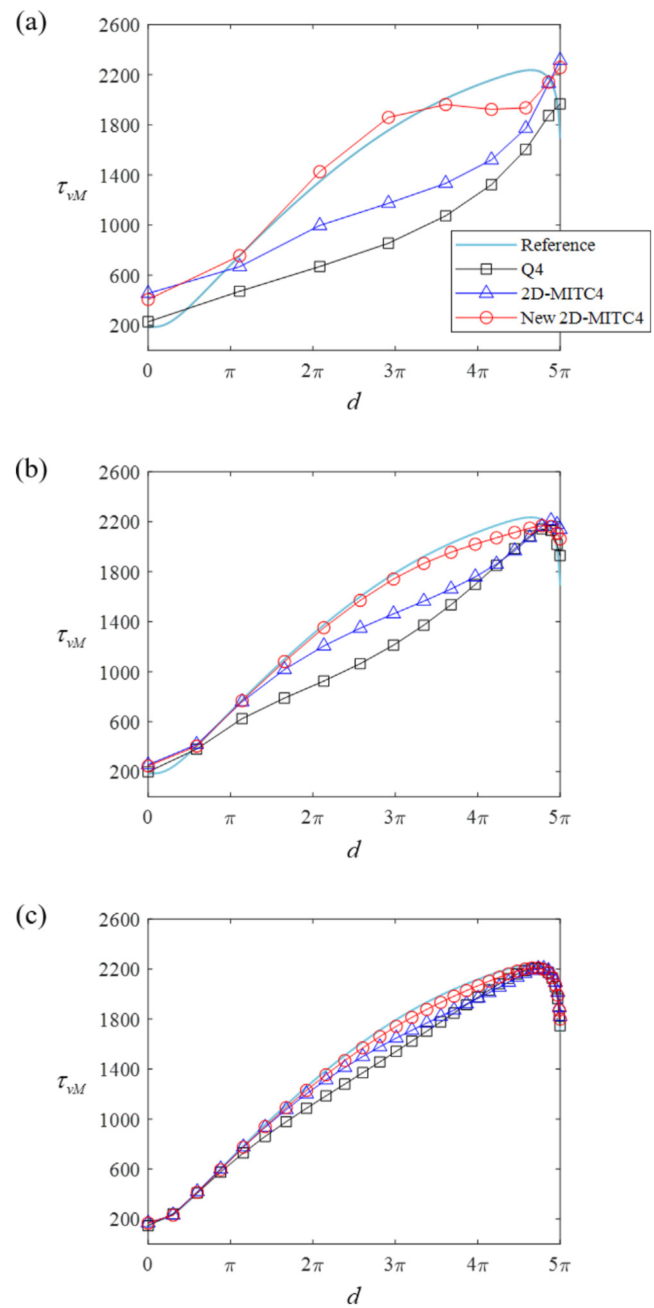

Fig. 22. Von mises stress (τ_{vM}) distributions along line *AB* of curved beam for (a) 8×8, (b) 16×16, and (c) 32×32 distorted meshes.

2.4. Strain-displacement relation and stiffness matrix

In vector and matrix forms, the assumed strain field in Eqs. (34)-(37) is transformed into the global Cartesian coordinates [1],

$$\varepsilon_{ij}^{AS} = \overset{\frown}{e}{}_{kl}^{AS}(\boldsymbol{i}_i \cdot \overset{\frown}{\mathbf{g}}{}^{k})(\boldsymbol{i}_j \cdot \overset{\frown}{\mathbf{g}}{}^{l}),\ \text{with}\ \boldsymbol{i}_1 = \boldsymbol{i}_x,\ \text{and}\ \boldsymbol{i}_2 = \boldsymbol{i}_y, \tag{38}$$

in which $\boldsymbol{i}_i$ denotes the global Cartesian base vectors. Note that $\varepsilon_{11}^{AS} = \varepsilon_{xx}^{AS}$, $\varepsilon_{22}^{AS} = \varepsilon_{yy}^{AS}$ and $\varepsilon_{12}^{AS} = \varepsilon_{xy}^{AS}$.

The relation between the assumed strain components in Eq. (38) and the nodal displacement vector is expressed as

$$\boldsymbol{e} = \mathbf{B}\mathbf{U}\ \text{with}\ \boldsymbol{e} = \begin{bmatrix} \varepsilon_{xx}^{AS} & \varepsilon_{yy}^{AS} & 2\varepsilon_{xy}^{AS} \end{bmatrix}^{\mathrm{T}} \text{and}\ \mathbf{U} = \begin{bmatrix} \mathbf{u}_1^{\mathrm{T}} & \mathbf{u}_2^{\mathrm{T}} & \mathbf{u}_3^{\mathrm{T}} & \mathbf{u}_4^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, \tag{39}$$

where $\mathbf{B}$ is the strain–displacement matrix, and $\mathbf{U}$ is the nodal displacement vector.

The stiffness matrix of the 2D-MITC4 element is obtained as

$$\mathbf{K} = \int_{V_e} \mathbf{B}^{\mathrm{T}}\mathbf{C}\mathbf{B}\, dV_e, \tag{40}$$

where V_e is the element volume and $\mathbf{C}$ is the material law matrix [1–3].

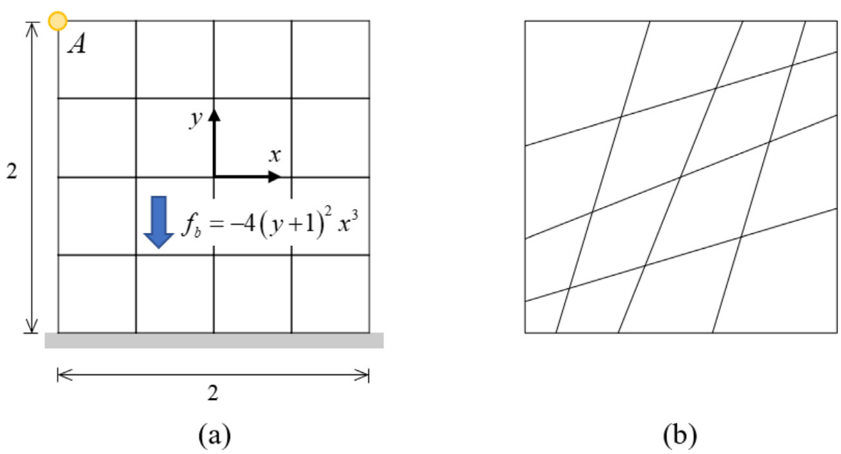

Fig. 23. Block under body force ($E = 2.0 \times 10^7$ and $\nu = 0.3$). (a) Regular mesh when $N = 4$. (b) Distorted mesh I when $N = 4$.

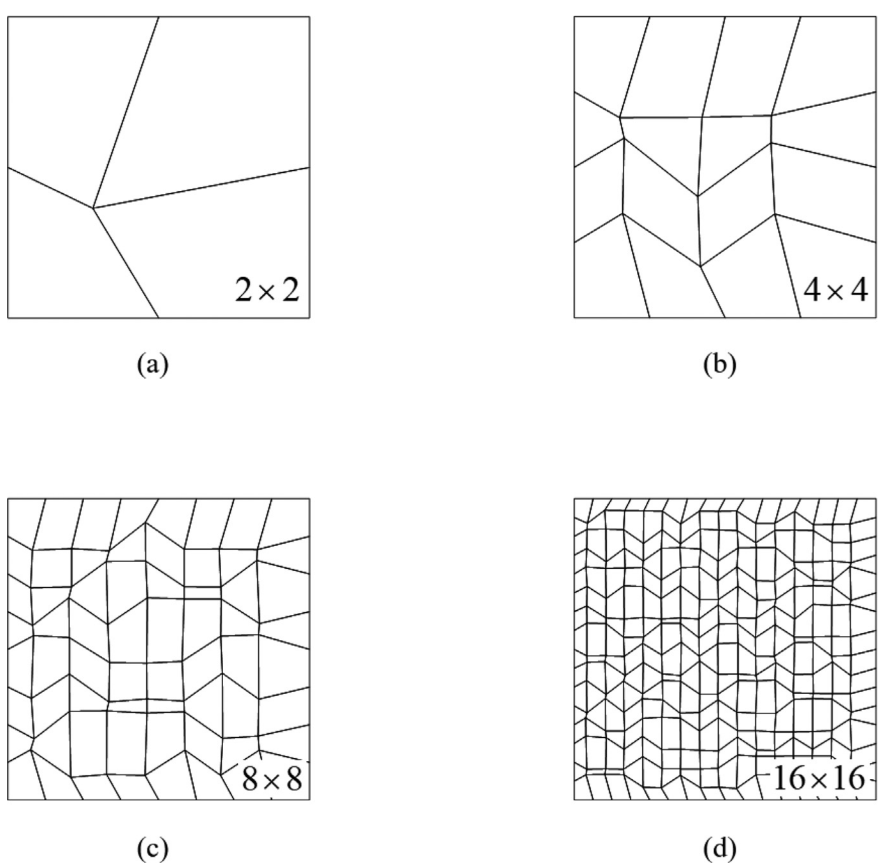

Fig. 24. Distorted mesh II when (a) $N = 2$, (b) $N = 2$, (c) $N = 8$ and (d) $N = 16$.

2.5. Alleviation of volumetric locking

To alleviate volumetric locking in the plane strain analysis of structures with nearly incompressible materials, the constant volumetric strain is assumed [1,23]. Note that this assumption can also improve the performance of the element in plane stress analysis [25].

The assumed strain field in Eq. (38) is decomposed into volumetric (denoted as "vol") and deviatoric (denoted as "dev") parts, as follows:

$$^{vol}e = \varepsilon_{xx}^{AS} + \varepsilon_{yy}^{AS} = {}^{vol}\,\mathbf{BU}, \tag{41}$$

$$^{dev}e_{ij} = \varepsilon_{ij}^{AS} - \frac{1}{2}\,{}^{vol}\,e\delta_{ij} = {}^{dev}\,\mathbf{B}_{ij}\mathbf{U}. \tag{42}$$

In Eq. (41), ε_{xx}^{AS} and ε_{yy}^{AS} are obtained at the element center ($r = s = 0$).

Then, the element stiffness matrix of the 2D-MITC4 element can be calculated as

$$\mathbf{K} = \int_{V_e} \left(\kappa + \frac{\eta}{3}G\right)\,{}^{vol}\mathbf{B}^{\mathrm{T}}\,{}^{vol}\mathbf{B}dV_e + \int_{V_e} {}^{dev}\,\mathbf{B}_{ij}^{\mathrm{T}}\,C_{ij}^{dev}\,\mathbf{B}_{ij}^{dev}\,dV_e, \tag{43}$$

where $\eta = \frac{(1+\nu)(1-4\nu)}{(1-\nu)(1-2\nu)}$ for the plane stress analysis, $\eta = 1$ for the plane strain analysis, κ and G are the bulk and shear modulus, respectively, and C_{ij}^{dev} denotes the material tensor for the deviatoric strain and stress ($C_{11}^{dev} = C_{22}^{dev} = 2G$ and $C_{12}^{dev} = C_{21}^{dev} = G$).

3. New 2D-MITC4 element

In this section, we introduce a geometry dependent Gauss integration scheme to enhance the bending behavior of the 2D-MITC4 element when the element is distorted.

3.1. Classification of distortion

The geometry distortion of a 4-node quadrilateral element can be expressed by the aspect ratio, skewness, and taper [4,45]. It is interesting to note that the geometry distortion due to skewness and taper can be represented using three characteristic vectors, $\mathbf{x}_r$, $\mathbf{x}_s$, and $\mathbf{x}_d$ as shown in Fig. 4**(a)**.

Using the two vectors $\mathbf{x}_r$ and $\mathbf{x}_s$, a parallelogram with blue dashed line can be constructed as shown in Fig. 4**(b)**. The skewness can be measured by angle θ and the skew angle is calculated as

$$\cos\theta = \frac{1}{|\widehat{\mathbf{g}}_r|\,|\widehat{\mathbf{g}}^r|} = \frac{1}{|\widehat{\mathbf{g}}_s|\,|\widehat{\mathbf{g}}^s|}. \tag{44}$$

In addition, the characteristic vector $\mathbf{x}_d$ representing the taper can be measured by α and β in Eq. (37).

$$\mathbf{x}_d = \alpha\mathbf{x}_r + \beta\mathbf{x}_s, \tag{45}$$

where α and β denote the tapers in the r- and s- directions, respectively; see Figs. 4**(c)** and **(d)**.

3.2. Geometry dependent Gauss integration

It is well known that the performance of finite elements can be improved by modifying the integration rule [10–14,40–42]. In most previous studies, a fixed integration rule was adopted without concern for element geometry. We here introduce a numerical scheme to adaptively adjust Gauss integration points according to the degree of element distortion to make the element performance less sensitive to its geometry distortion.

When evaluating the stiffness matrix, the standard 2×2 Gauss integration in Fig. 5**(a)** is employed in general

$$\mathbf{K} = t\sum_{i=1}^{2}\sum_{j=1}^{2} w_i w_j \mathbf{F}(\zeta_i, \zeta_j) \text{ with } \zeta_1 = 1/\sqrt{3},\ \zeta_2 = -1/\sqrt{3},\ w_1 = w_2 = 1, \tag{46}$$

in which t is the element thickness, and ζ_i and w_i denote the integration positions and the corresponding weight factors for the two-point Gauss integration, respectively.

In Eq. (46),

$$\mathbf{F}(r,s) = \mathbf{B}^{\mathrm{T}}\mathbf{CB}\det(\boldsymbol{J}) \text{ for the 2D-MITC4 element} \tag{47}$$

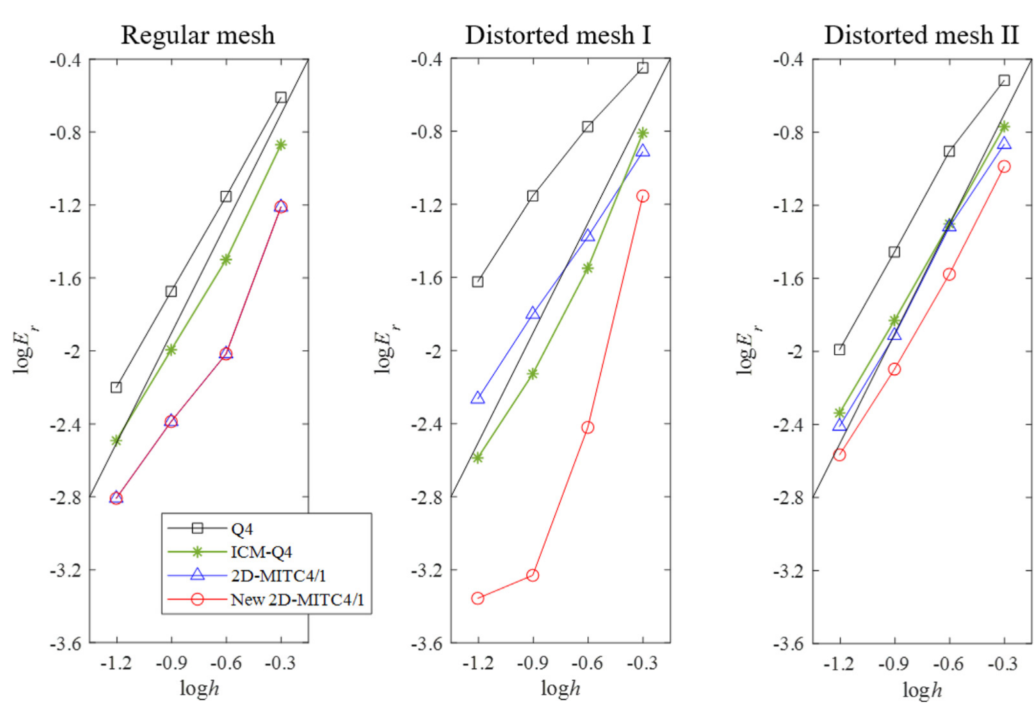

Fig. 25. Convergence curves for block under body force. The bold line denotes the optimal convergence rate.

and

$$\mathbf{F}(r,s) = \left[\left(\kappa + \frac{\eta}{3} G \right) {}^{vol}\mathbf{B}^{\mathrm{T}}\, {vol}\mathbf{B} + {}^{dev}\, \mathbf{B}_{ij}^{\mathrm{T}} C_{ij}^{dev}\, {dev}\mathbf{B}_{ij} \right] \det(\boldsymbol{J})$$

for the 2D-MITC4/1 element, (48)

With this standard integration rule, the original 2D-MITC4 and MITC4/1 elements reveal overly stiff bending behavior when the element is distorted. This issue will be presented in Section 5.

This stiffening effect arising with geometry distortion can be alleviated by moving the integration points toward the element center as

$$\mathbf{K} \approx t \sum_{i=1}^{2} \sum_{i=1}^{2} w_i w_j \mathbf{F}\left(\hat{\zeta}_i, \hat{\zeta}_j\right) \quad \text{with} \quad \hat{\zeta}_i = \mu \zeta_i, \tag{49}$$

where μ is an adjusting parameter smaller than 1.0, as shown in Fig. 5**(b)**.

To show that adjusted integration points can alleviate overly stiff bending behavior of the distorted element, the eigen analysis of the stiffness matrix of a single 2D-MITC4 element shown in Fig. 6 is carried out. The plane stress condition is considered with Poisson's ratio $\nu = 0.3$.

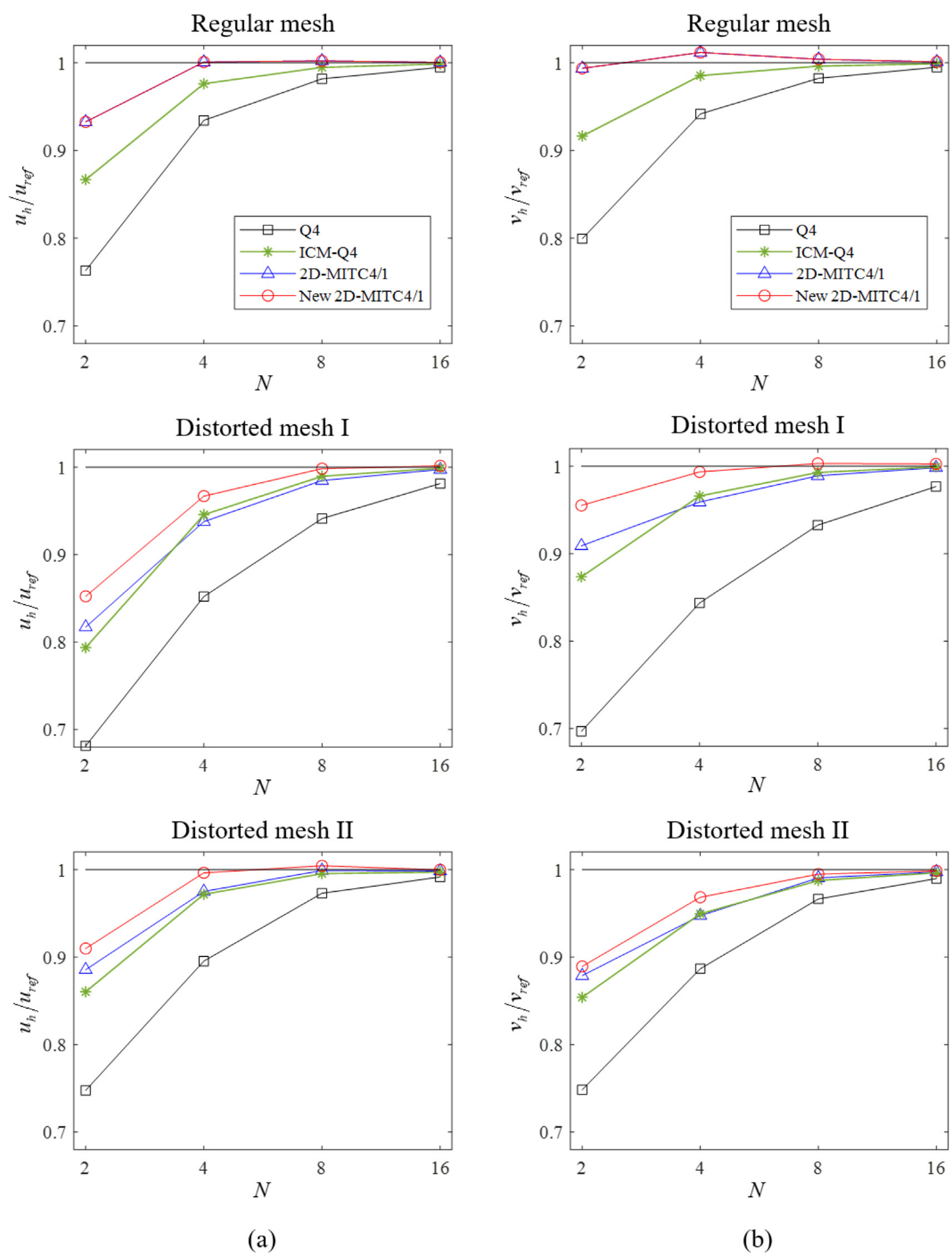

Fig. 26. Normalized (a) horizontal (u_h/u_{ref}) and (b) vertical (v_h/v_{ref}) displacements at point A in block under body force.

Table 1 shows eigenvalues normalized by Young's modulus according to the adjusting parameter μ. The standard 2×2 Gauss integration is applied with the parameter $\mu = 1$. The case of $\mu = 0$ corresponds to the reduced integration, for which the eigenvalues for the two bending modes become zero and thus spurious zero energy modes occur.

Fig. 7 shows the normalized eigenvalues of each deformation mode according to the adjusting parameter. The eigenvalues for the constant strain modes (modes 6, 7 and 8) are almost unaffected, but the eigenvalues corresponding to the bending modes (modes 4 and 5) rapidly decrease as the adjusting parameter become smaller. Therefore, it is possible to adaptively reduce the element bending stiffness by changing the integration points.

With this observation, we propose the following practical requirements to design a function $\mu = f(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4)$ for the adjusting parameter.

- $\mu = 1$ for undistorted element geometries (Standard Gauss integration is used.)
- $\mu \leqslant 1$ when element geometry is distorted
- No spurious zero energy mode: $\mu > 0$
- Ideally optimal convergence for bending problems
- Same value for quadrilaterals in similarity: $\mu_{\square ABCD} = \mu_{\square EFGH}$ when $\square ABCD \sim \square EFGH$

A number of functions for the adjusting parameter satisfying the above requirements can be devised, and we here propose the following function

$$\mu = \cos^2(\theta) = \frac{1}{|\widehat{\mathbf{g}}_r|^2 \, |\widehat{\mathbf{g}}^r|^2} = \frac{1}{|\widehat{\mathbf{g}}_s|^2 \, |\widehat{\mathbf{g}}^s|^2}, \quad (50)$$

in which θ denotes the skew angle defined in Eq. (44). Fig. 8 shows the relationship between the adjusting parameter μ and the skew angle θ.

The same adjusting parameter is also employed to calculate strain and stress. Strain at (r', s') is calculated by entering $(\mu r', \mu s')$ into Eqs. (39), (41) and (42) instead of (r', s'), and the corresponding stress is obtained through the material law.

In this study, the total Lagrangian formulation is employed for geometric nonlinear analysis and thus the adjusting parameter is obtained from the initial element geometry. If the updated Lagrangian formulation is adopted, the adjusting parameter could be updated.

4. Basic numerical tests

We perform three basic numerical tests (patch, zero energy mode, and isotropy tests) for the new 2D-MITC4 element [1–3,43]. In this section, the new 2D-MITC4 element refers to both the new 2D-MITC4 and 2D-MITC4/1 elements.

Patch tests for stretching and shearing are carried out using the mesh shown in Fig. 9**(a)** [1]. The minimum number of constraints are given to prevent rigid body motions. The new 2D-MITC4 element produces analytically correct constant stress fields. That is, the element passes the patch tests.

In the zero energy mode test, a single 2D element without support should have only three zero energy modes corresponding to the correct rigid body motions: two translations and one rotation [25]. Various element geometries including the geometry shown in Fig. 9**(b)** are considered and the new 2D-MITC4 element passes the zero energy mode test.

The behavior of a finite element should not be affected by the node numbering sequence [43]. That is, spatially isotropic behavior is an essential requirement for finite elements. The finite element model in Fig. 9**(c)** is solved. The new 2D-MITC4 element provides identical results regardless of the node numbering sequence and thus it passes the isotropy test. Of course, the element passes the test in other finite element models.

5. Numerical examples

To demonstrate the performance of the new 2D-MITC4 element, we solve several numerical examples: a cantilever beam, a cantilever beam modeled using two elements with distortion parameter, Cook's skew beam, a curved beam, a block under a body force, a block subjected to a distributed compression force, and a column under a compressive load. The standard 4-node quadrilateral element (Q4), the original 2D-MITC4 element (2D-MITC4), and the incompatible modes element (ICM-Q4) are considered for comparison with the new 2D-MITC4 element (new 2D-MITC4). The 2D-

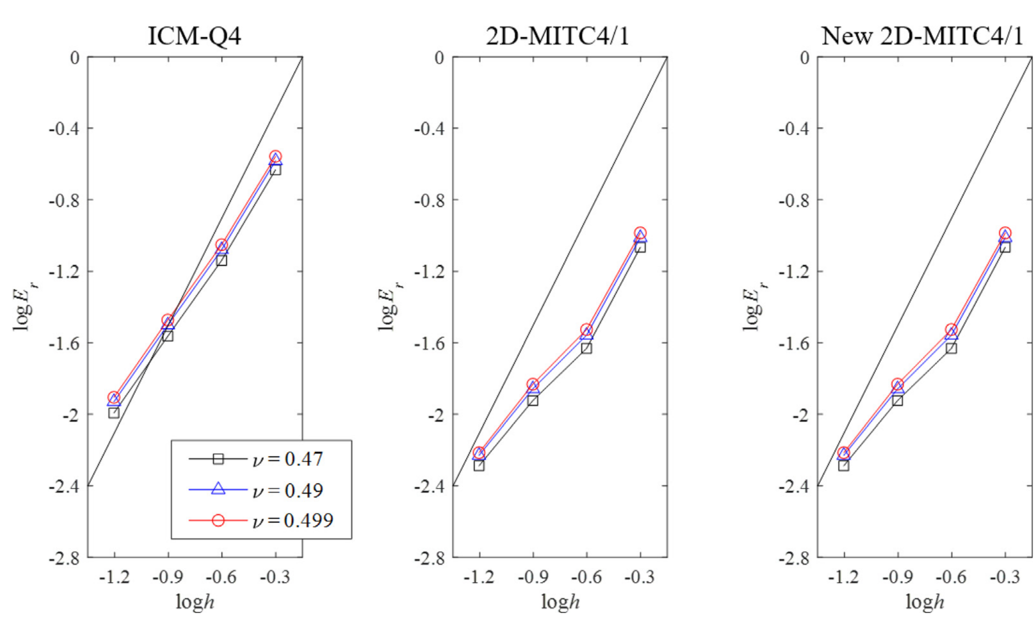

Fig. 27. Convergence curves for block under body force with regular mesh. Nearly incompressible material properties are considered. The bold line denotes the optimal convergence rate.

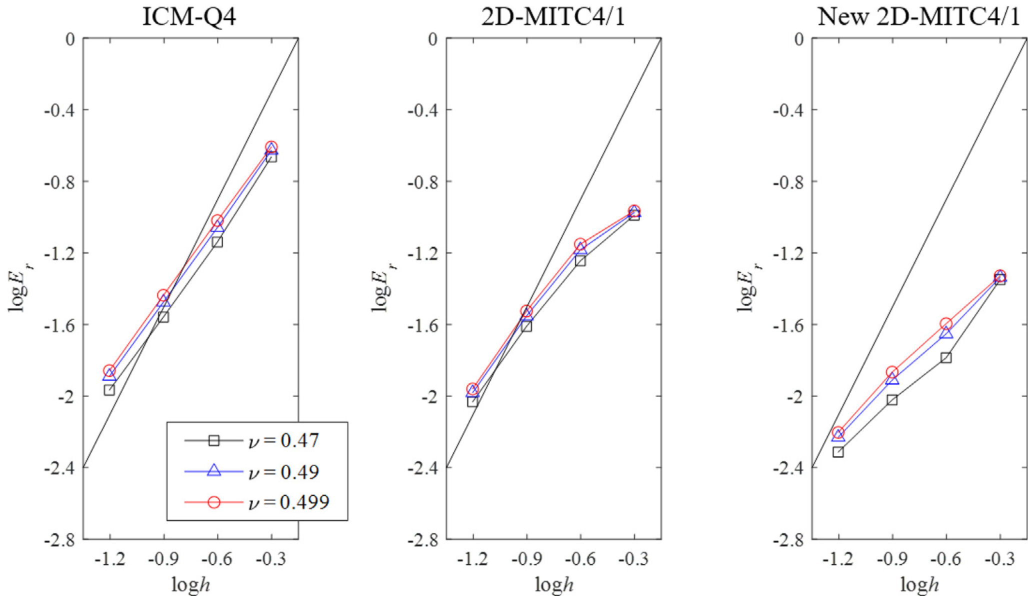

Fig. 28. Convergence curves for block under body force with distorted mesh I. Nearly incompressible material properties are considered. The bold line denotes the optimal convergence rate.

MITC4/1 and new 2D-MITC4/1 elements denote that the treatment of volumetric locking is applied.

To measure the predictive capability of the 4-node finite elements considered, we evaluate the displacements and stresses at specific locations. The convergence of the relative error in strain energy E_r is also plotted

$$E_r = \frac{|E_{ref} - E_h|}{E_{ref}}, \tag{51}$$

where E_{ref} and E_h are the strain energies calculated from the reference and finite element solutions, respectively. The optimal convergence for the 4-node elements is $E_r \cong ch^2$, where c is a constant and h denotes the element size [1]. The reference solutions are either analytical solutions or sufficiently converged numerical solutions obtained using 9-node quadrilateral elements.

$N \times N$ regular and distorted meshes are generated for the finite element models. The mesh patterns are first constructed in the square domain shown in Fig. 10 and, then, linearly mapped to the problem domain under consideration. In the distorted mesh pattern, each edge is divided into the following ratio: $L_1 : L_2 : L_3 : \ldots\ldots L_N = 1 : 2 : 3 : \ldots\ldots N$. In this study, the thickness is given as 1.0 in all 2D solid problems and $h = 1/N$ for the element size.

5.1. Cantilever beam

Let us consider a cantilever beam subjected to shearing force $P = 40$ at its free end, as shown in Fig. 11 [25]. The plane stress condition is considered with Young's modulus $E = 3.0 \times 10^4$ and Poisson's ratio $\nu = 0$. The cantilever beam is modeled using four elements in regular and distorted meshes.

The relative errors in the vertical displacement at point A and the xx-component of stress at point B for both regular and distorted meshes are listed in Tables 2 and 3, respectively. For the regular mesh, the ICM-Q4 and the original and new 2D-MITC4 elements give excellent solutions. For the distorted mesh, the new 2D-MITC4 element shows much better accuracy than the original 2D-MITC4 element. Its performance is very similar to that of the ICM-Q4 element.

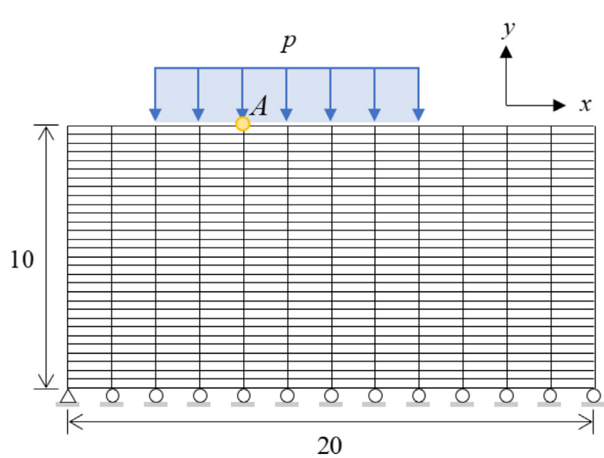

Fig. 29. Block subjected to distributed compression force with 12×30 mesh ($E = 1.0 \times 10^3$, $\nu = 0.3$ and $p_{\max} = -98$).

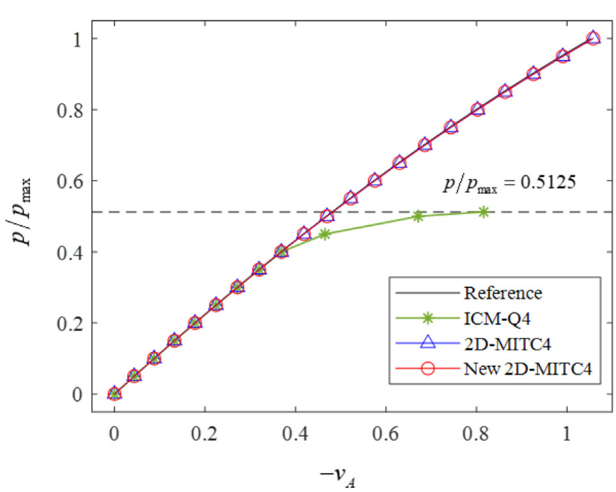

Fig. 30. Load-displacement curves at point A in block subjected to distributed compression force. The deformed shapes at load step $p/p_{\max} = 0.5125$ are given in Fig. 31.

5.2. Cantilever beam modeled using two elements with distortion parameter

Solving the cantilever beam modeled using two elements with a distortion parameter, as shown in Fig. 12 [5], the mesh distortion sensitivity is investigated. The plane stress condition is considered with Young's modulus $E = 1500$ and Poisson's ratio $\nu = 0.25$. The left end is fixed and a bending moment $M = 2000$ is applied at the right end.

As the distortion parameter e changes from 0 to 4.9, the element distortion becomes more severe. The normalized vertical displacement at point A is shown in Fig. 13. The new 2D-MITC4 and 2D-MITC4/1 elements are less affected by the element distortion than is the original 2D-MITC4 and 2D-MITC4/1 elements.

5.3. Cook's skew beam

Cook's skew beam problem is considered as shown in Fig. 14**(a)** [2,43]. The left edge of the structure is fixed, and a distributed shearing force $f_s = 1/16$ (force per length) is applied along its right edge. We use the plane stress condition with Young's modulus $E = 1.0$ and Poisson's ratio $\nu = 1/3$. Solutions for both regular and distorted meshes are obtained with $N \times N$ meshes (N = 2, 4, 8, and 16). The regular and distorted mesh patterns used for the problem are depicted in Figs. 14**(b)** and **(c)**, respectively.

The convergence curves for strain energy are shown in Fig. 15. The convergences of both the normalized horizontal and vertical displacements at point A are shown in Figs. 16**(a)** and **(b)**, respectively. The relative errors in the horizontal and vertical displacements are given in Tables 4 and 5, respectively. The shear stress distributions obtained using 8×8 distorted meshes are shown in Fig. 17. The reference solutions for displacements and strain energy are obtained using a 64×64 mesh of 9-node quadrilateral elements. The new 2D-MITC4/1 element gives significantly more accurate results than does the original 2D-MITC4/1 element, in particular for distorted meshes.

5.4. Curved beam

We consider the curved beam problem as shown in Fig. 18**(a)** [25]. The curved beam is clamped along its bottom and subjected to a distributed shearing force $f_s = 120$ (force per length) along its free tip. The plane stress condition with $E = 1.0 \times 10^3$ and Poisson's ratio $\nu = 0$ is adopted. The convergence behavior is studied considering both regular and distorted meshes of $N \times N$ elements (N = 2, 4, 8, and 16). The regular and distorted mesh patterns used for the problem are given in Figs. 18**(b)** and **(c)**, respectively.

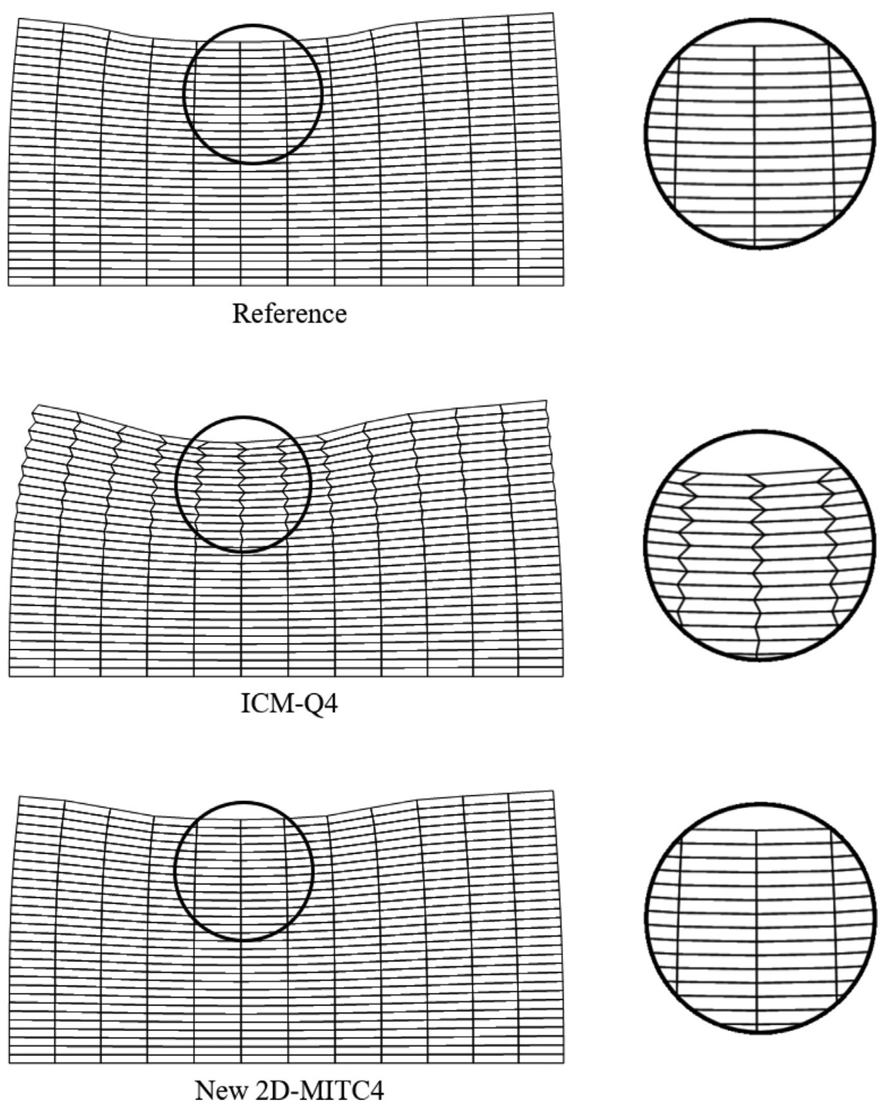

Fig. 31. Deformed configurations of block subjected to distributed compression force at load step $p/p_{\max} = 0.5125$ with displacements magnified two times.

The convergence in strain energy is given in Fig. 19. The normalized vertical displacements at point A are shown in Fig. 20. The von Mises stress distributions obtained using the 16×16 distorted mesh are given in Fig. 21. In addition, the von Mises stress (averaged at the nodes) along the line AB are depicted in Fig. 22. Note that the results of the 2D-MITC4 and 2D-MITC4/1 elements are identical since Poisson's ratio is zero. The reference solutions are obtained by using a 64×64 mesh of 9-node quadrilateral elements. When the regular meshes are used, all elements except for the Q4 element show similarly good performance. However, for the distorted mesh case, the new 2D-MITC4 element shows much better predictive capability than that of the original 2D-MITC4 element.

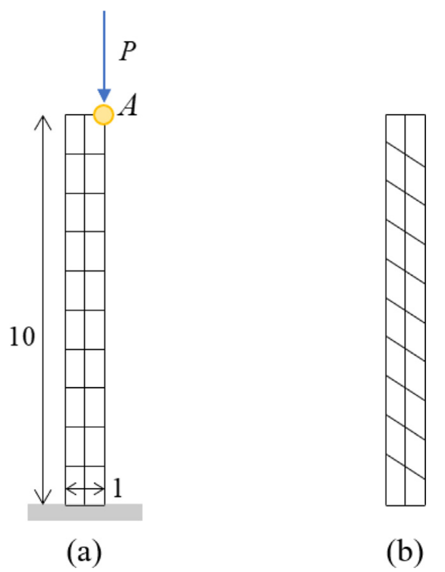

Fig. 32. Column under compressive load ($E = 1.0 \times 10^6$, $\nu = 0$ and $P_{max} = 4.5 \times 10^3$) modeled with (a) 2×10 regular mesh and (b) 2×10 distorted mesh.

5.5. Block under body force

A square box is clamped along its bottom and subjected to a body force $f_s = -4(y+1)^2x^3$ (force per area), as shown in Fig. 23 [23]. The plane strain condition is employed with Young's modulus $E = 2.0 \times 10^7$ and Poisson's ratio $\nu = 0.3$. Solutions are obtained with $N \times N$ regular and distorted meshes ($N =$ 2, 4, 8, and 16), see Fig. 23. Two distorted mesh patterns are considered: distorted mesh I in Fig. 23(**b**) and distorted mesh II in Fig. 24.

The convergences in strain energy are given in Fig. 25. The normalized horizontal and vertical displacements at point A are shown in Fig. 26. The reference solutions are obtained using a 64×64 mesh of 9-node quadrilateral elements. For the regular mesh pattern, the 2D-MITC4/1 and new 2D-MITC4/1 elements provide the same results, but the new 2D-MITC4/1 element gives better solutions than the original 2D-MITC4/1 element when the distorted mesh patterns are considered.

We then perform plane strain analysis considering nearly incompressible materials with three different Poisson's ratios $\nu =$ 0.47, 0.49 and 0.499. The convergence curves of the relative errors in strain energy for regular and distorted meshes are given in Figs. 27 and 28, respectively. The new 2D-MITC4/1 element provides the most accurate solutions, especially in distorted meshes.

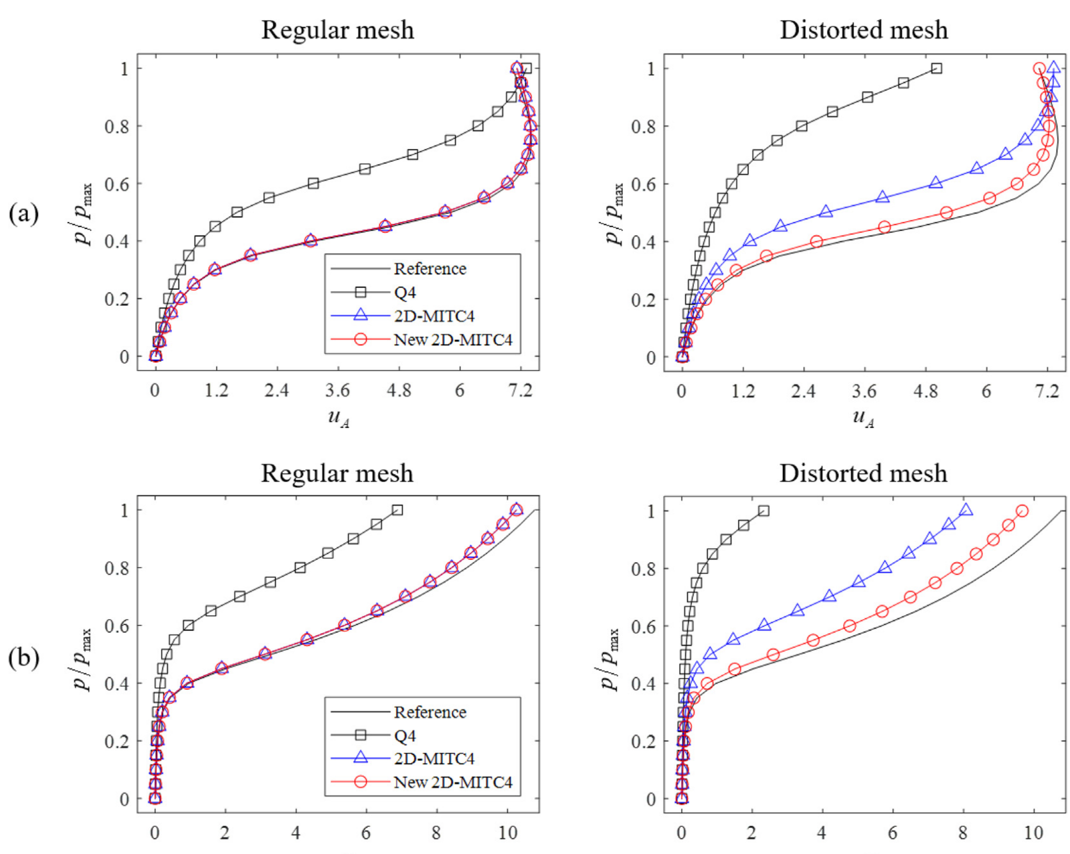

Fig. 33. Load-displacement curves of column under compressive load for regular and distorted meshes. (a) Horizontal and (b) vertical displacements.

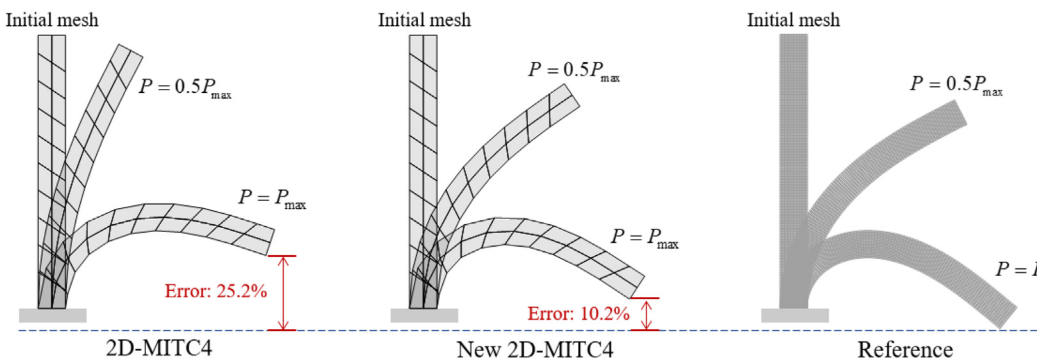

Fig. 34. Deformed configurations of column under compressive load at initial, intermediate, and final load steps, obtained using 2×10 distorted meshes. Reference solution is obtained by 20×100 mesh of 9-node quadrilateral elements.

5.6. Block subjected to distributed compression force

We solve the geometric nonlinear problem to investigate whether the new 2D-MITC4 element shows spurious instabilities that appear in the ICM-Q4 element [22,23]. A block under a distributed compression force $p_{max} = -98$ (force per length) is considered, as shown in Fig. 29. The plane stress condition with Young's modulus $E = 1.0 \times 10^3$ and Poisson's ratio $\nu = 0.3$ is adopted and a 12×30 mesh is used. We consider three elements: ICM-Q4, original 2D-MITC4, and new 2D-MITC4 elements.

Fig. 30 shows load–displacement curves representing the vertical displacement at point *A* according to the compression force. The deformed configurations at the load step $p/p_{max} = 0.5125$ are given in Fig. 31. The 12×30 mesh of 9-node quadrilateral elements is used to obtain the reference solutions. The original and new 2D-MITC4 elements closely follow the reference path over the entire set of load steps, without any spurious instability.

5.7. Column under compressive load

Let us consider the geometric nonlinear analysis of a column under an eccentric load, as shown in Fig. 32 [44]. The bottom of the column is fixed and the eccentric load $P_{max} = 4.5 \times 10^3$ is applied at the right top of the column. The plane stress condition with Young's modulus $E = 1.0 \times 10^6$ and Poisson's ratio $\nu = 0$ is used with a 2×10 mesh. The distorted mesh is given in Fig. 32**(b)**.

Figs. 33**(a)** and **(b)** show load–displacement curves representing the horizontal and vertical displacement, respectively, at point *A*, according to the compression force. The deformed configurations at the initial, intermediate, and final load steps are given in Fig. 34. The 20×100 mesh of 9-node quadrilateral elements is used to obtain the reference solutions. For the regular meshes, the original and new 2D-MITC4 elements provide similarly accurate results, but the new 2D-MITC4 element performs much better than the original 2D-MITC4 element in distorted meshes.

6. Concluding remarks

In this study, a new 2D-MITC4 element (including a new 2D-MITC4/1 element) is developed using geometry dependent Gauss integration. By utilizing the skew angle of the element, the integration points were adjusted in accordance with the geometry distortion. We also presented a simplified assumed strain field that provides the same performance as the original 2D-MITC4 element but has a much simpler formulation. The new 2D-MITC4 element passed all basic tests including the isotropy, patch, and zero energy mode tests and revealed no numerical instabilities in geometric nonlinear analysis. In various numerical examples, the new 2D-MITC4 element showed better convergence behavior than that of the original 2D-MITC4 element, especially in distorted meshes. That is, the original 2D-MITC4 element was successfully improved.

In future work, the concept of geometry dependent Gauss integration can be applied to other solid and structural elements. It will also be valuable to extend the simplified assumed strain field to a 4-node quadrilateral shell element to alleviate membrane locking in a much simpler way than is employed in the MITC4 + shell element.

Data availability

The authors do not have permission to share data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Bathe K.J. Finite element procedures. Prentice Hall; 1996, 2nd ed. K.J. Bathe, Watertown, MA; 2014 and Higher Education Press, China; 2016.
[2] Cook RD. Concepts and applications of finite element analysis. New York: John Wiley & Sons; 2007.
[3] Hughes TJR. The finite element method: linear static and dynamic finite element analysis. Mineola, NY: Dover Publications; 2000.
[4] Lee NS, Bathe KJ. Effects of element distortions on the performance of isoparametric elements. Int J Numer Methods Eng 1993;36:3553–76.
[5] Jung J, Jun H, Lee PS. Self-updated four-node finite element using deep learning. Comput Mech 2022;69:23–44.
[6] MacNeal RH. A theorem regarding the locking of tapered four-noded membrane elements. Int J Numer Methods Eng 1987;24:1793–9.
[7] Panasz P, Wisniewski K, Turska E. Reduction of mesh distortion effects for nine-node elements using corrected shape functions. Finite Elem Anal Des 2013;66:83–95.
[8] MacNeal RH. Toward a defect-free four-noded membrane element. Finite Elem Anal Des 1989;5:31–7.
[9] Jung J, Yoon K, Lee PS. Deep learned finite elements. Comput Methods Appl Mech Eng 2020;372:113401.
[10] Barlow J. More on optimal stress points – reduced integration, element distortions and error estimation. Int J Numer Methods Eng 1989;28:1487–504.
[11] Zienkiewicz OC, Taylor RL, Too JM. Reduced integration technique in general analysis of plates and shells. Int J Numer Methods Eng 1971;3:275–90.

[12] Belytschko T, Tsay CS. A stabilization procedure for the quadrilateral plate element with one-point quadrature. Int J Numer Methods Eng 1983;19:405–19.
[13] Malkus DS, Hughes TJR. Mixed finite element methods – reduced and selective integration techniques: A unification of concepts. Comput Methods Appl Mech Eng 1978;15:63–81.
[14] Griffiths DV, Mustoe GGW. Selective reduced integration of four-node plane element in closed form. J Eng Mech 1995;121:725–9.
[15] Simo JC, Rifai MS. A class of mixed assumed strain methods and the method of incompatible modes. Int J Numer Methods Eng 1990;29:1595–638.
[16] Bischoff M, Ramm E, Braess D. A class of equivalent enhanced assumed strain and hybrid stress finite elements. Comput Mech 1999;22:443–9.
[17] Kožar I, Rukavina T, Ibrahimbegović A. Method of incompatible modes – overview and application. J Croat Assoc Civ Eng 2018;70:19–29.
[18] Wilson EL, Taylor RL, Doherty WP, Ghaboussi J. Incompatible displacement models. INC.: ACADEMIC PRESS; 1973.
[19] Taylor RL, Beresford PJ, Wilson EL. A non-conforming element for stress analysis. Int J Numer Methods Eng 1976;10:1211–9.
[20] Bischoff M, Romero I. A generalization of the method of incompatible modes. Int J Numer Methods Eng 2007;69:1851–68.
[21] Wilson EL, Ibrahimbegović A. Use of incompatible displacement modes for the calculation of element stiffnesses or stresses. Finite Elem Anal Des 1990;7:229–41.
[22] Sussman T, Bathe KJ. Spurious modes in geometrically nonlinear small displacement finite elements with incompatible modes. Comput Struct 2014;140:14–22.
[23] Lee C, Kim S, Lee PS. The strain-smoothed 4-node quadrilateral finite element. Comput Methods Appl Mech Eng 2021;373:113481.
[24] Ko Y, Bathe KJ. A new 8-node element for analysis of three-dimensional solids. Comput Struct 2018;202:85–104.
[25] Ko Y, Lee PS, Bathe KJ. A new 4-node MITC element for analysis of two-dimensional solids and its formulation in a shell element. Comput Struct 2017;192:34–49.
[26] Dvorkin EN, Bathe KJ. A continuum mechanics based four-node shell element for general non-linear analysis. Eng Comput 1984;1:77–88.
[27] Bucalem ML, Bathe KJ. Higher-order MITC general shell elements. Int J Numer Methods Eng 1993;36:3729–54.
[28] Bathe KJ, Lee PS, Hiller JF. Towards improving the MITC9 shell element. Comput Struct 2003;81:477–89.
[29] Lee PS, Bathe KJ. Development of MITC isotropic triangular shell finite elements. Comput Struct 2004;82:945–62.
[30] Beirão da Veiga L, Chapelle D, Paris SI. Towards improving the MITC6 triangular shell element. Comput Struct 2007;85:1589–610.
[31] Lee Y, Lee PS, Bathe KJ. The MITC3+ shell element and its performance. Comput Struct 2014;138:12–23.
[32] Ko Y, Lee PS, Bathe KJ. A new MITC4+ shell element. Comput Struct 2017;182:404–18.
[33] Ko Y, Lee PS. A 6-node triangular solid-shell element for linear and nonlinear analysis. Int J Numer Methods Eng 2017;111:1203–30.
[34] Ko Y, Lee PS, Bathe KJ. The MITC4+ shell element in geometric nonlinear analysis. Comput Struct 2017;185:1–14.
[35] Ko Y, Lee Y, Lee PS, Bathe KJ. Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Comput Struct 2017;193:187–206.
[36] Jun H, Yoon K, Lee PS, Bathe KJ. The MITC3+ shell element enriched in membrane displacements by interpolation covers. Comput Methods Appl Mech Eng 2018;337:458–80.
[37] Wisniewski K, Turska E. Improved nine-node shell element MITC9i with reduced distortion sensitivity. Comput Mech 2018;62:499–523.
[38] Dvorkin EN, Vassolo SI. A quadrilateral 2-D finite element based on mixed interpolation of tensorial components. Eng Comput 1989;6:217–24.
[39] Bathe KJ, Dvorkin EN. A formulation of general shell elements – the use of mixed interpolation of tensorial components. Int J Numer Methods Eng 1986;22:697–722.
[40] Belytschko T, Engelmann BE. On flexurally superconvergent four-node quadrilaterals. Comput Struct 1987;25:909–18.
[41] Stander N, Wilson EL. A 4-node quadrilateral membrane element with in-plane vertex rotations and modified reduced quadrature. Eng Comput 1989;6:266–71.
[42] Reddy BD, Küssner M. The four-noded quadrilateral with a 2×1 integration rule: Application to plates and other problems. Comput Methods Appl Mech Eng 1997;149:101–12.
[43] Lee C, Lee PS. A new strain smoothing method for triangular and tetrahedral finite elements. Comput Methods Appl Mech Eng 2018;341:939–55.
[44] Lee C, Lee DH, Lee PS. The strain-smoothed MITC3+ shell element in nonlinear analysis. Comput Struct 2022;266:106768.
[45] MacNeal RH, Harder RL. A proposed standard set of problems to test finite element accuracy. Finite Elem Anal Des 1985;1:3–20.
[46] Robinson J. CRE method of element testing and the Jacobian shape parameters. Eng Comput 1987;4:113–8.